Create Job plans in Maximo from Asset maintenance manuals.

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1. Background

Asset-intensive industries often rely on **OEM(Original Equipment Manufacturer)-provided maintenance manuals** to guide equipment upkeep. These manuals typically contain unstructured, narrative-style instructions that are difficult to operationalize directly within enterprise systems like **IBM Maximo**.

Traditionally, maintenance engineers manually interpret these manuals and input tasks into Maximo as job plans—a process that is **time-consuming, error-prone, and non-scalable**. This bottleneck delays maintenance planning, introduces inconsistencies across teams, and increases the risk of non-compliance with OEM standards.

To overcome these challenges, this use case proposes an **AI-powered solution** that automates the conversion of maintenance manual content into **structured Maximo job plans**.

Using IBM Watsonx capabilities:

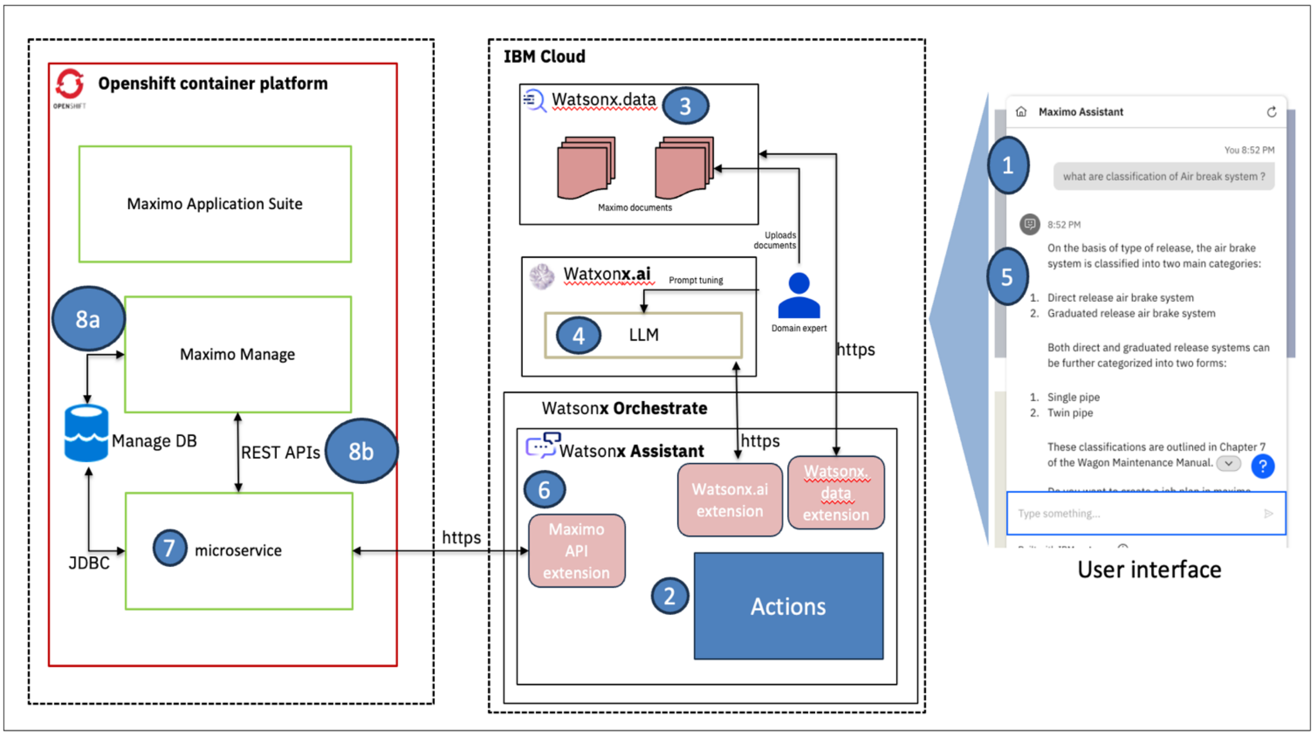
- **Watsonx.ai** is leveraged to interpret and extract task sequences from unstructured PDF manuals using large language models (LLMs).
- The extracted steps are transformed into a JSON structure compatible with Maximo's job plan schema.
- These plans are then ingested into Maximo via APIs, enabling **automated creation and updates of job plans**.
- **Watsonx.data** (with Milvus) supports intelligent retrieval and indexing of manual content for RAG (Retrieval Augmented Generation) scenarios.

This automated pipeline ensures:

- **Standardized job plan generation**
- **Reduced manual effort**
- **Faster turnaround for preventive and corrective maintenance planning**
- **Alignment with OEM-specified procedures**

By streamlining this process, your maintenance team can scale efficiently, improve data integrity, and minimize equipment downtime

2. Architecture Diagram



3. Technologies used

Services	Purpose
Watsonx.ai Studio	To generate structured responses using LLMs
Watsonx.ai Runtime	To host and run foundation models
Watsonx Assistant	For building conversational search applications
Watsonx.Data (Milvus DB)	Storing and retrieving vectorized document data
Maximo	Asset Management Tool

4. Benefits Milvus Database

We are extracting and generating **job plans from asset maintenance manuals** using AI. These manuals are often large, unstructured documents with complex procedures scattered across many pages. To accurately retrieve relevant maintenance steps in real time, we need a **high-performance vector database** that can:

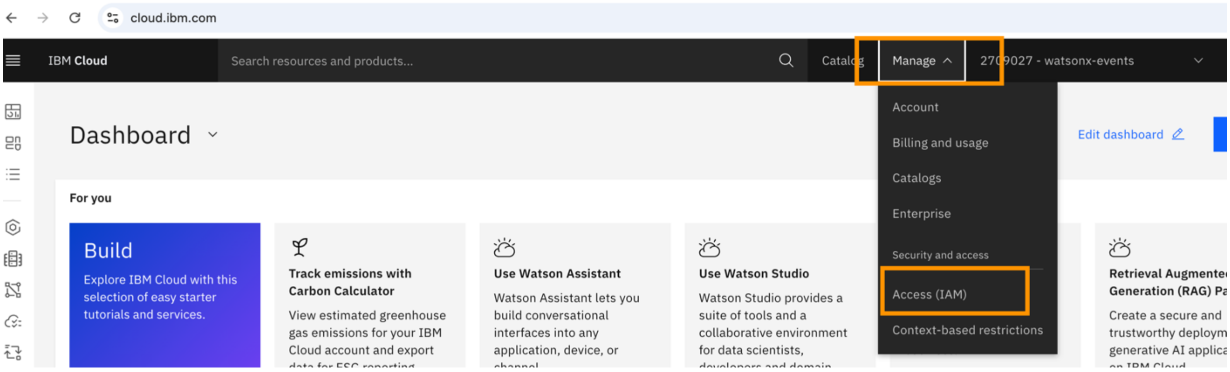
Feature	Why It Matters in This Use Case
🔍 **Vector Similarity Search**	Enables retrieval of the most relevant content chunks from manuals based on semantic similarity to the user's question or task.
⚡ **High Performance**	Designed for large-scale vector data, Milvus ensures fast retrieval even with thousands of embedded manual pages.
🧩 **RAG Enablement**	Powers the **Retrieval Augmented Generation (RAG)** workflow by feeding only the most relevant context to Watsonx.ai for accurate and grounded responses.
📁 **Unstructured Data Indexing**	Stores and indexes large volumes of text embeddings from PDFs, maintenance guides, and documents.
☁️ **Seamless Integration**	Works well with **Watsonx.data** and is supported within the IBM Cloud environment via infrastructure manager.

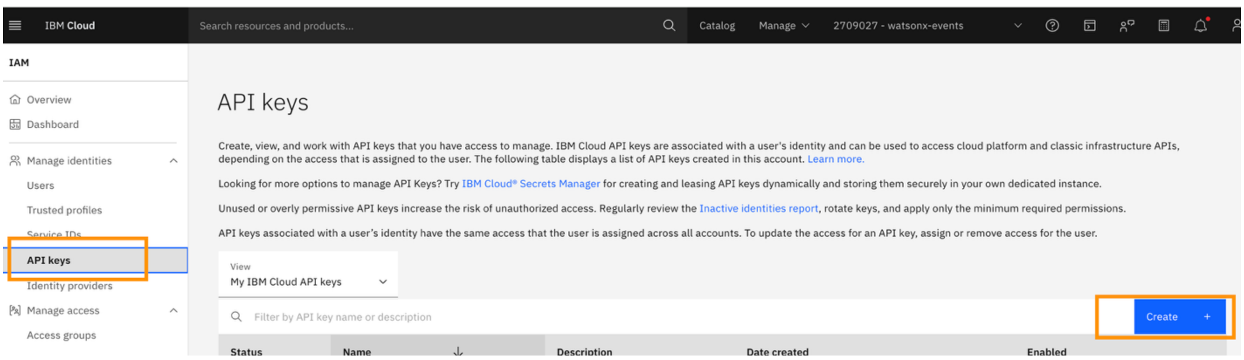
5. Create API key

Why to Create API key: An API key is required to securely authenticate and authorize communication between services in your Watsonx and Maximo environment.

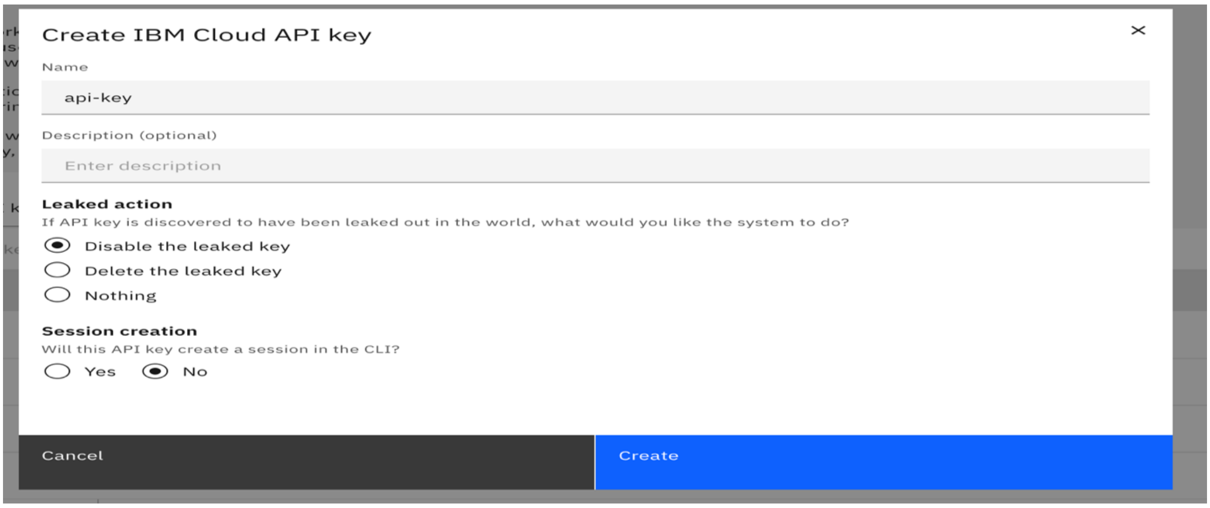
- Always keep API keys confidential.
- Rotate them periodically.
- Never hard-code them directly into source code (use .env or secrets managers instead).

5.1 Go to *IBM Cloud* dashboard →click on *Manage* →click on *Access (IAM)* →then click on *API Keys* on the left side → click on *Create*





5.2 Provide *Name* and *Description* → Click on *Create*→ Copy & download the API key for future reference



6. Watsonx.data Configurations Details

6.1 Copy the below connection details provided by the instructor.

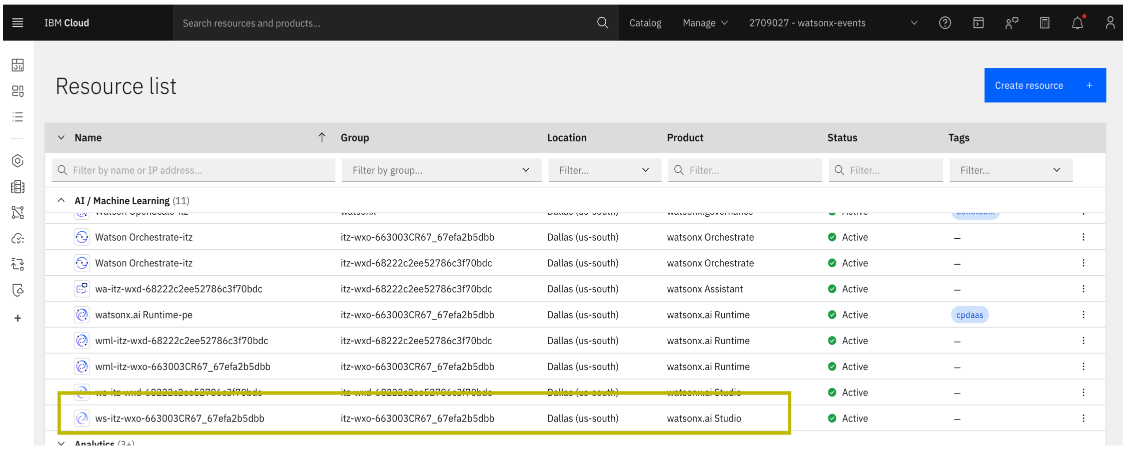
GRPC host: <host name>

GRPC port: <port>

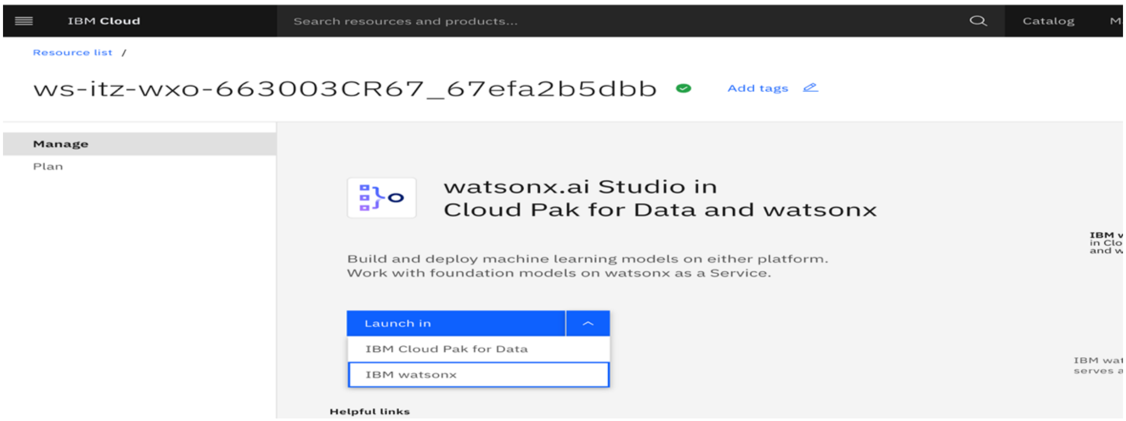
7. Watsonx.AI Configurations

7.1 Launch watsonx.ai

Click on *Resource list* from left side panel in *IBM Cloud* →Click on ***watsonx.ai Studio*** under ***AI / Machine Learning*** resource. (highlighted in image)

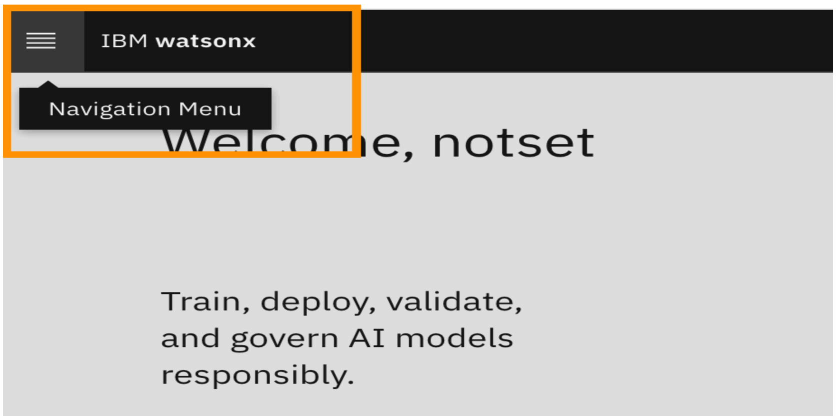


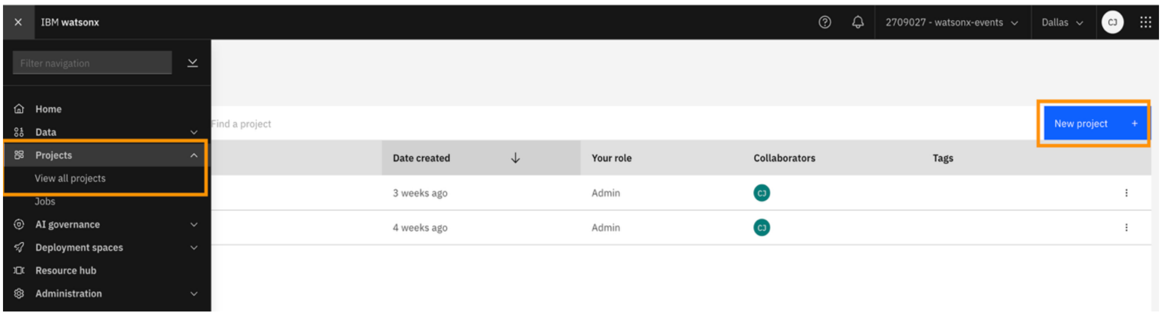
7.2 Select Launch in ***IBM watsonx***



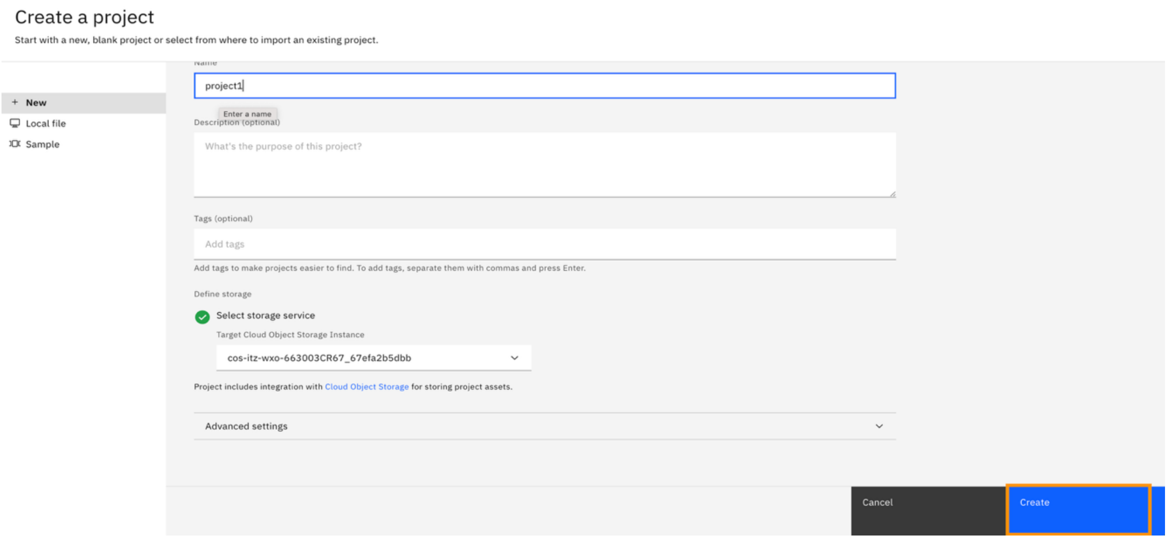
7.3 Create New project

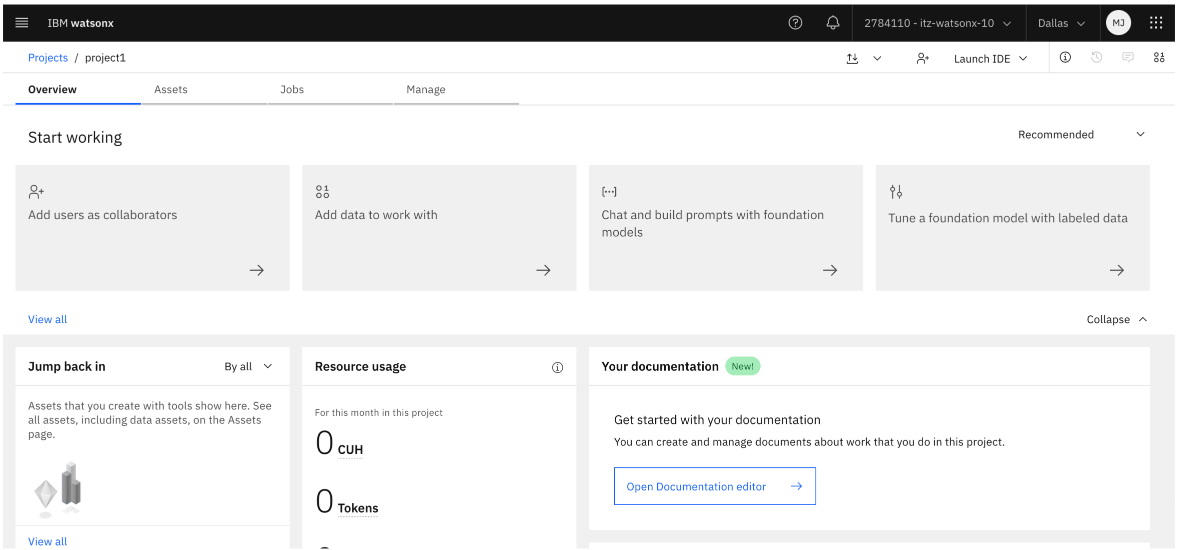
Click on *Navigation Menu* at top left side →Click on *Projects*→Click on *View all projects* → Click on *New project.*



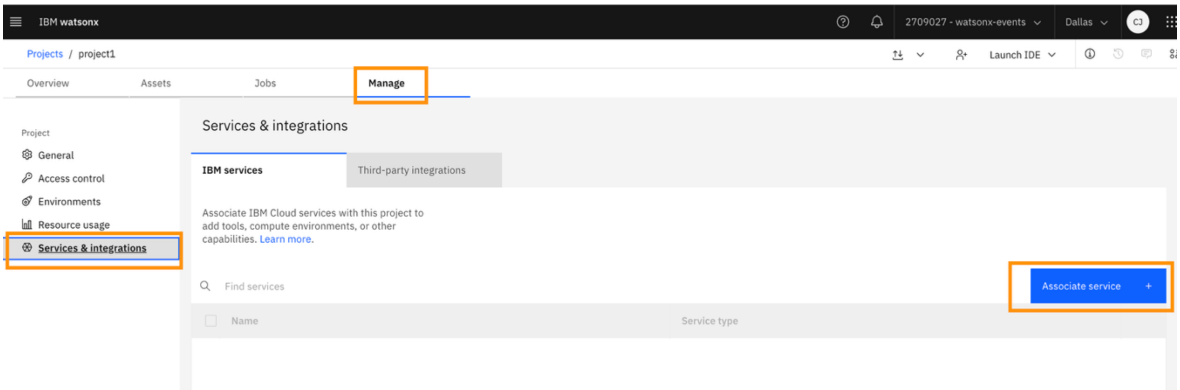


7.4 Provide *Name* and *Select storage service* from drop down →Click on *Create*.

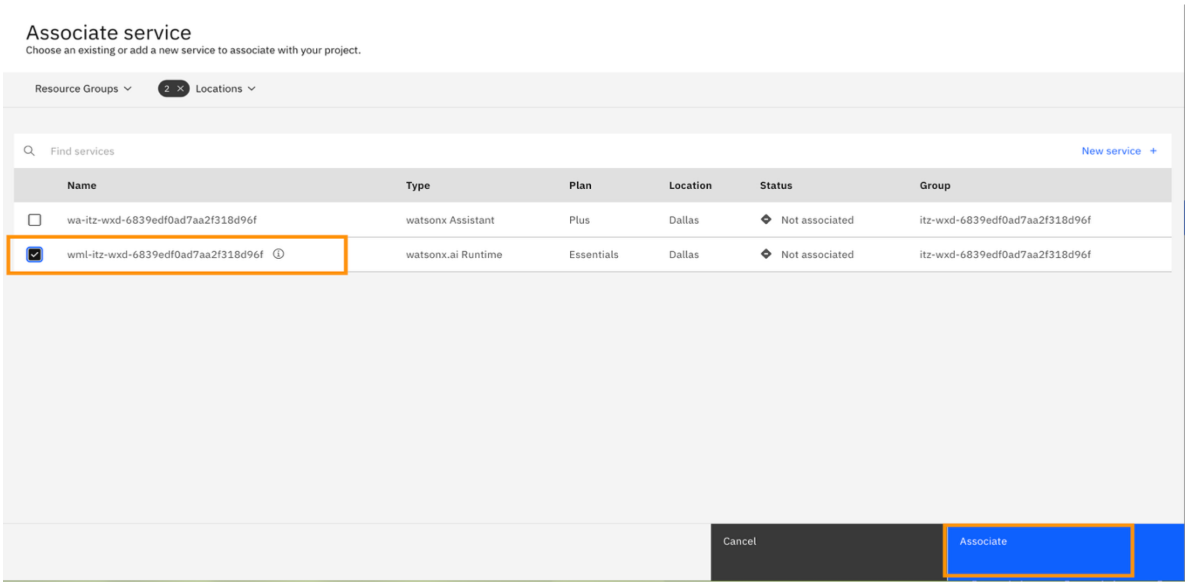




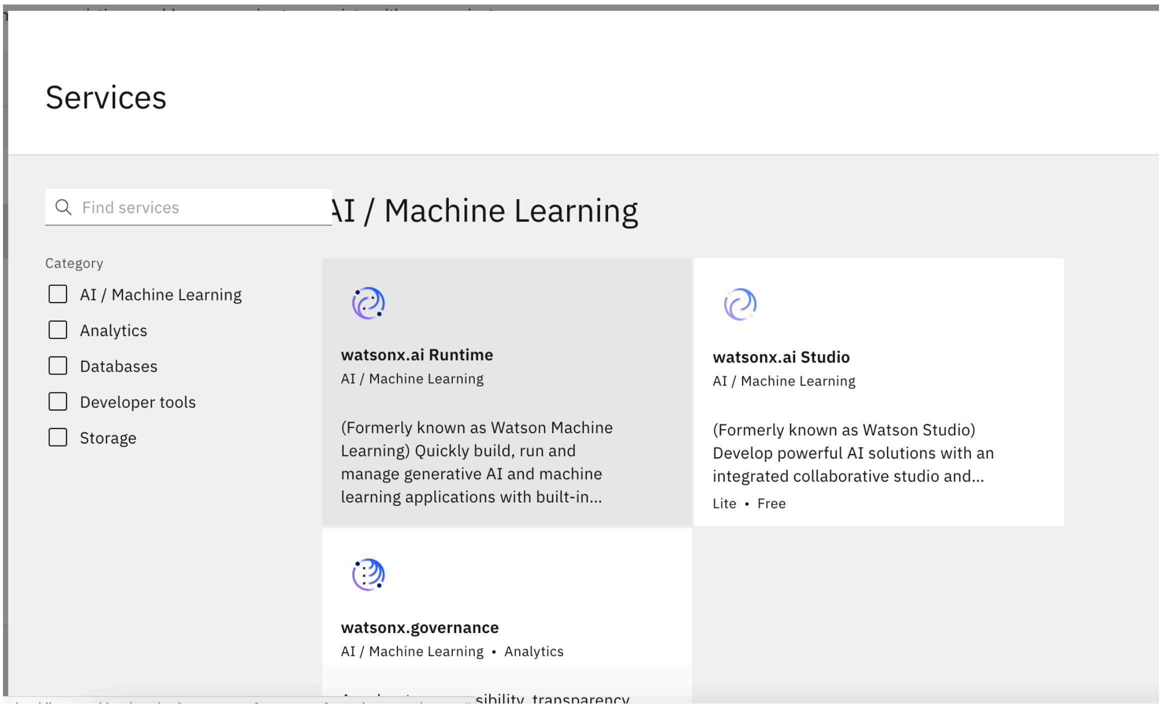
7.5 Once project created, Navigate to *Manage* tab → select *Services & integrations* (left side panel) → click on *Associate service* (highlighted in image)



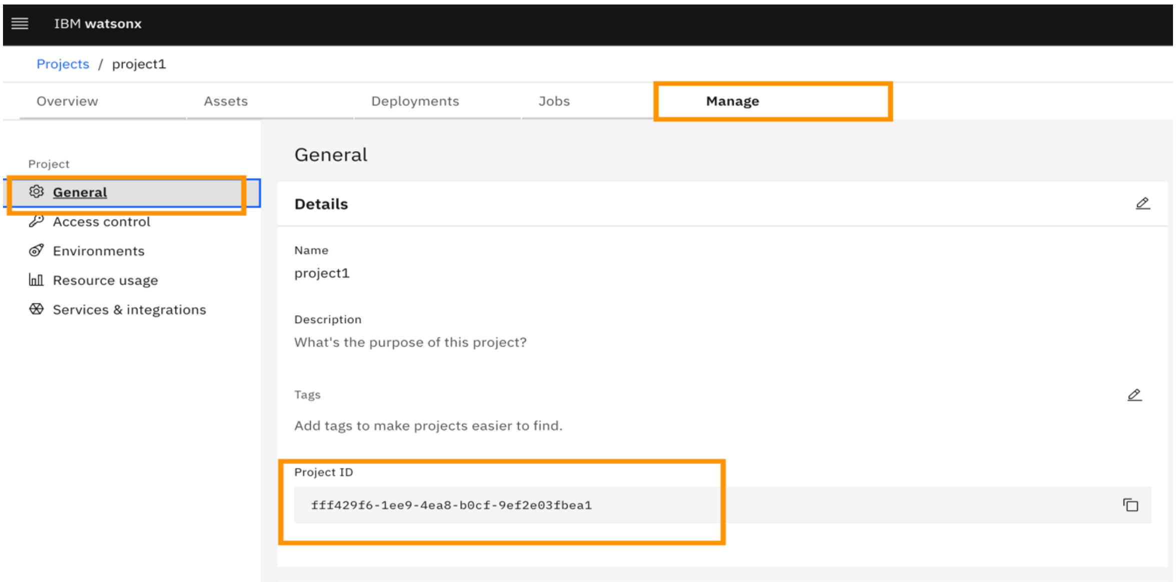
7.6 Select service name starts with wml(ex: wml-xxx-xxx-xxxxxxx) → click on *Associate*.



Note: if wml-xxx service does not display. Then click on New Service + icon and then select watsonx.ai Runtime as shown below and click on create.



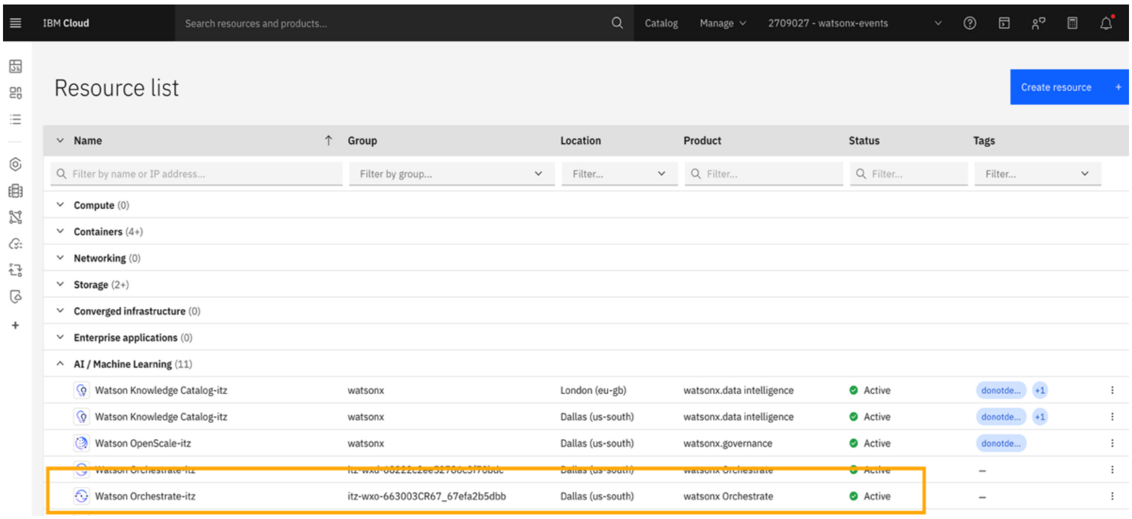
7.7 Again on *Manage* tab →Click on *General* →Copy and Save *Project ID* for the future use



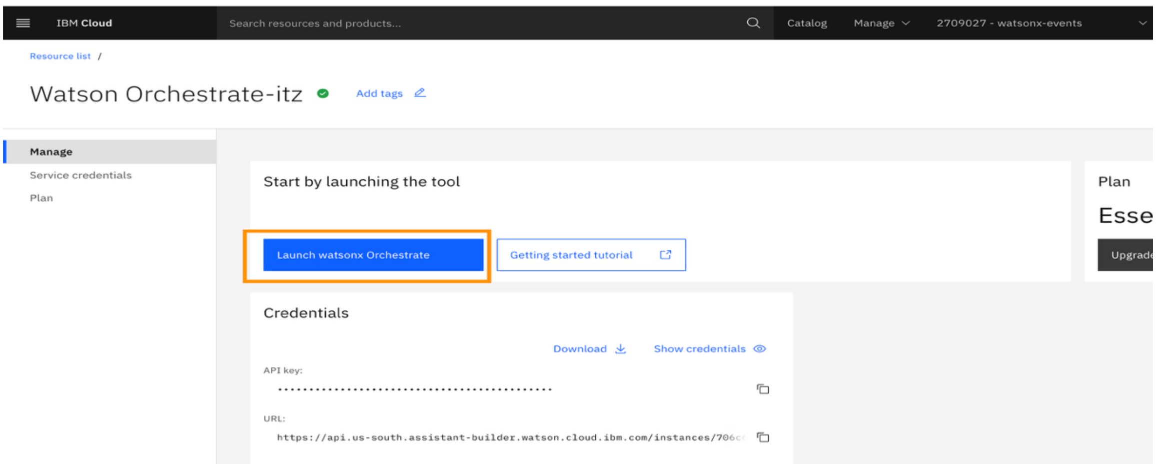
8. Watsonx Assistant Configurations

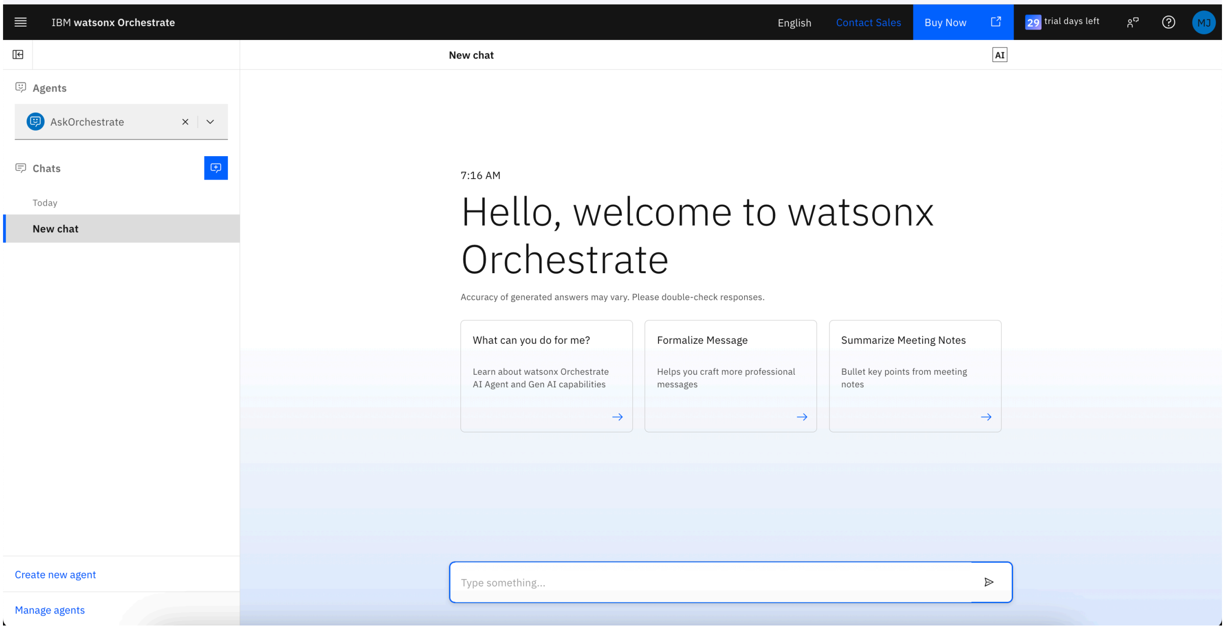
8.1 Launch watsonx assistance from orchestrate

Go to Resource list in *IBM Cloud* →expand *AI/Machine Learning* →Click on *Watson Orchestrate-* (highlighted in image)

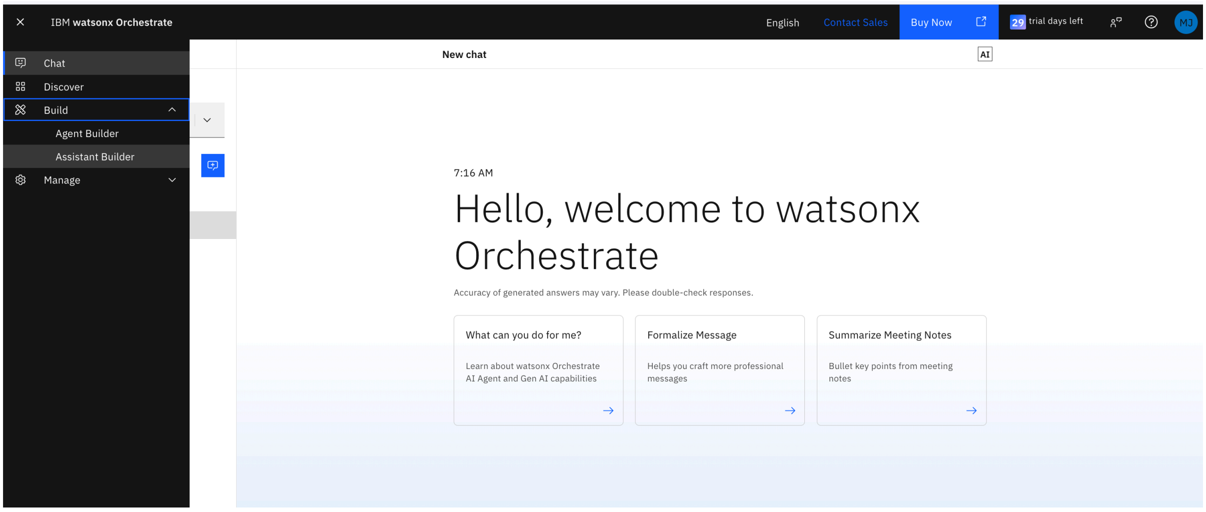


8.2 Click on *Launch watsonx Orchestrate*



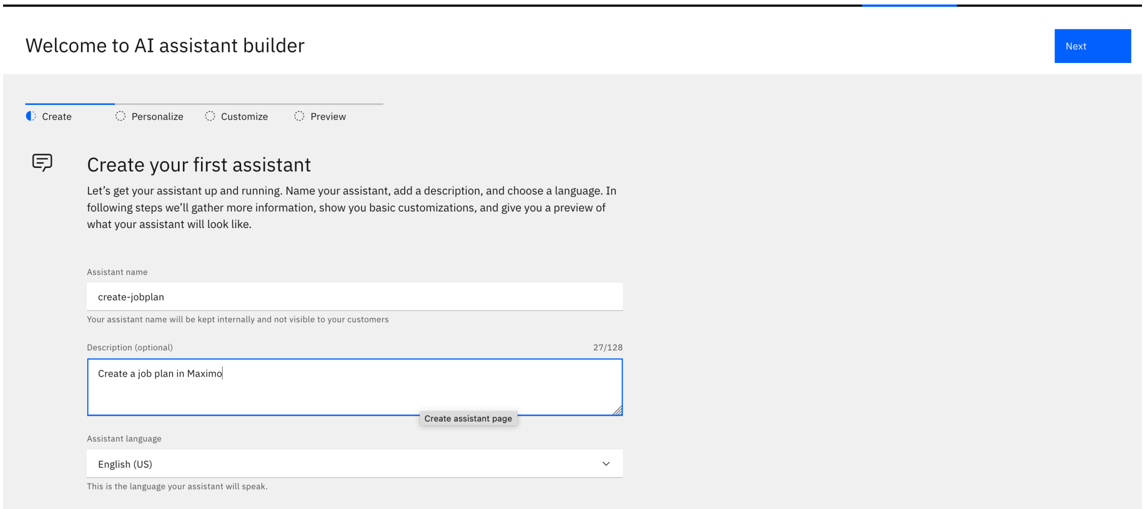


8.3 Click on left menu, Expand *Build* and select *Assistant Builder* option



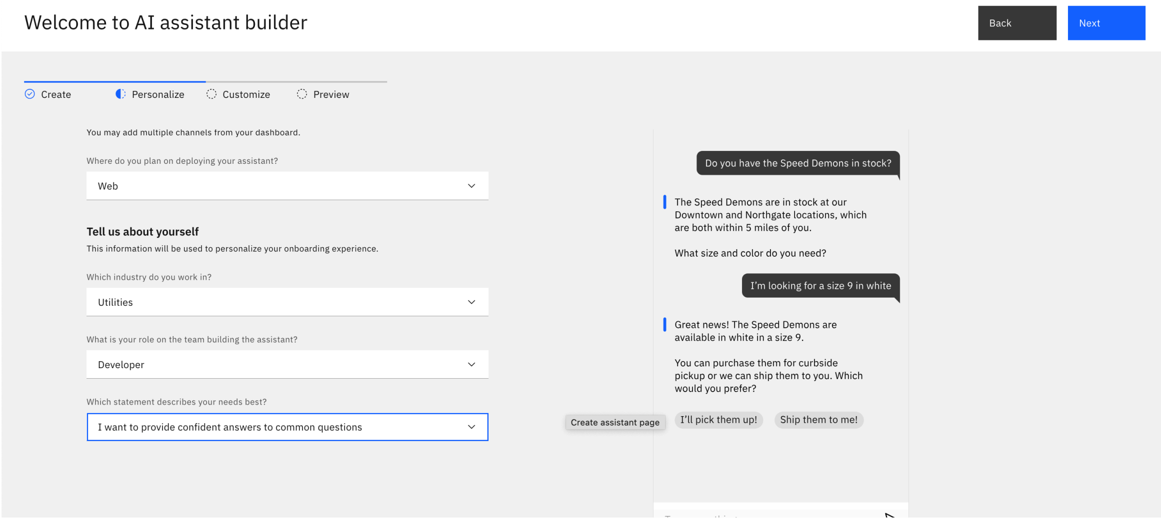
8.4 Enter the *Assistant Name* and optional Description and click on Next.

Note: Enter your initial name with Assistant Name, like Create-jobplan-XX

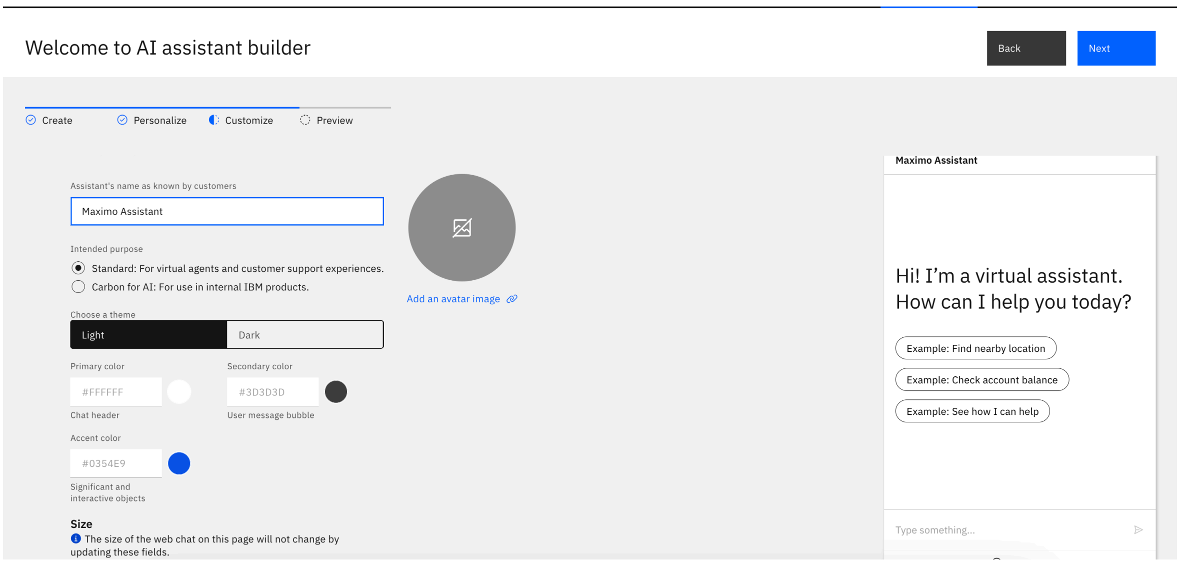


8.5 Select the options as below to *personalize the Assistant.*

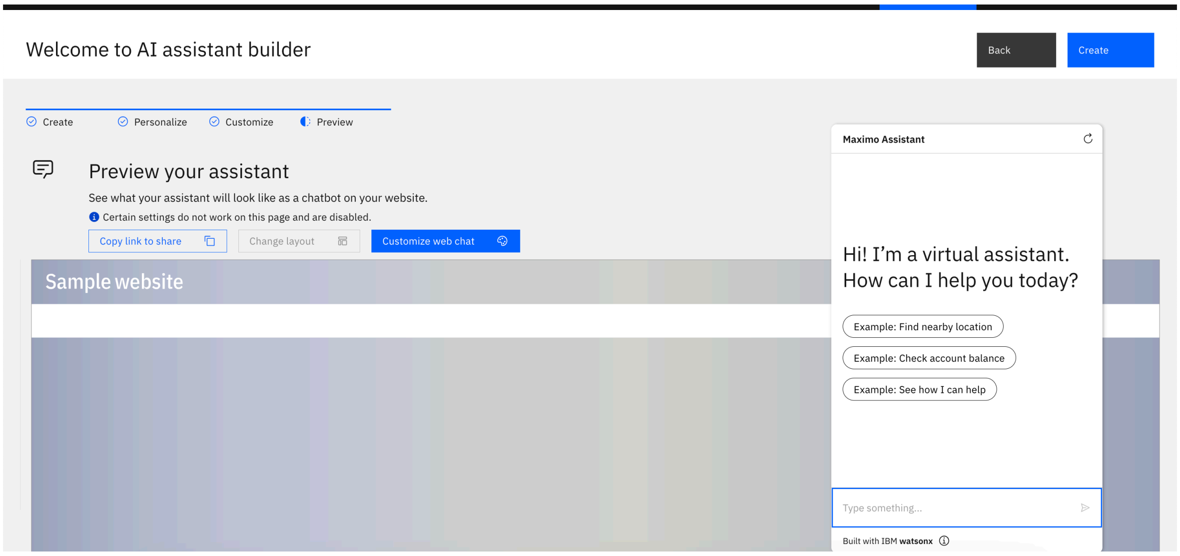
Note : You can select different values based on your requirement.

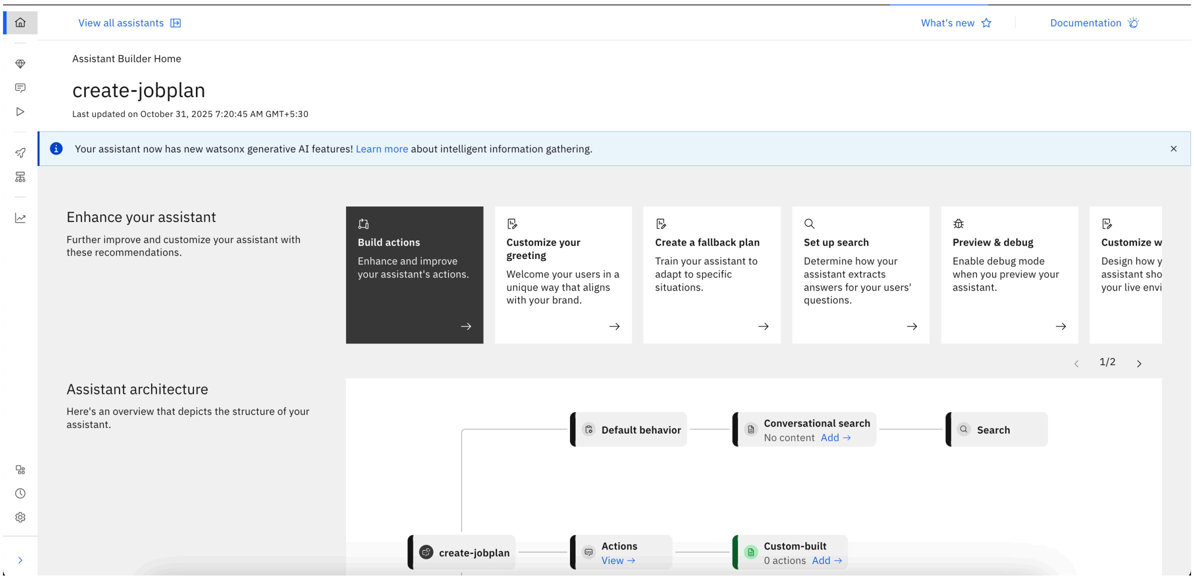


8.6 Enter Assistant name as *Maximo Assistant* and feel free to select the different theme if you wish to.



8.7 Click on Create button.

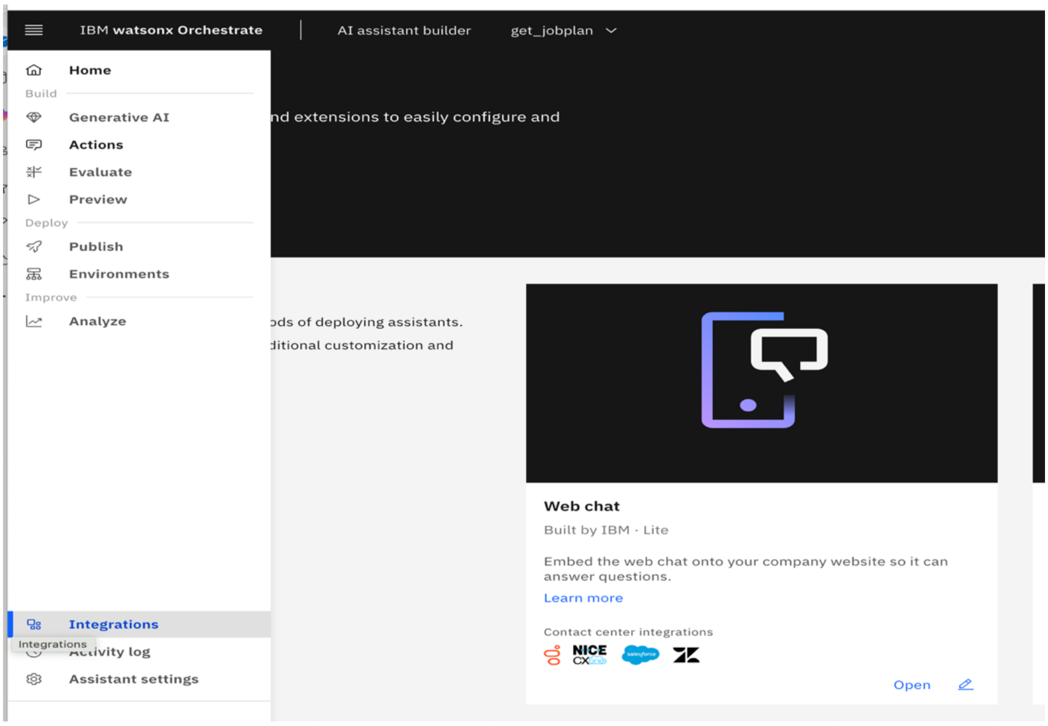




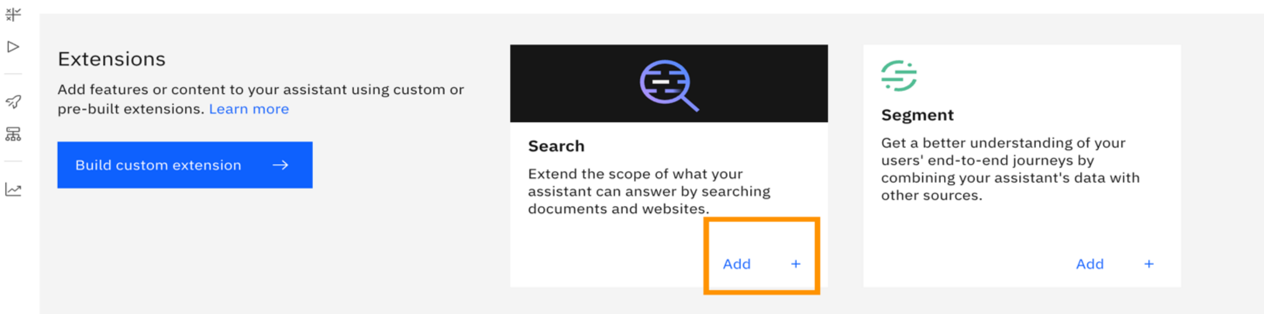
9. Adding Custom Extensions

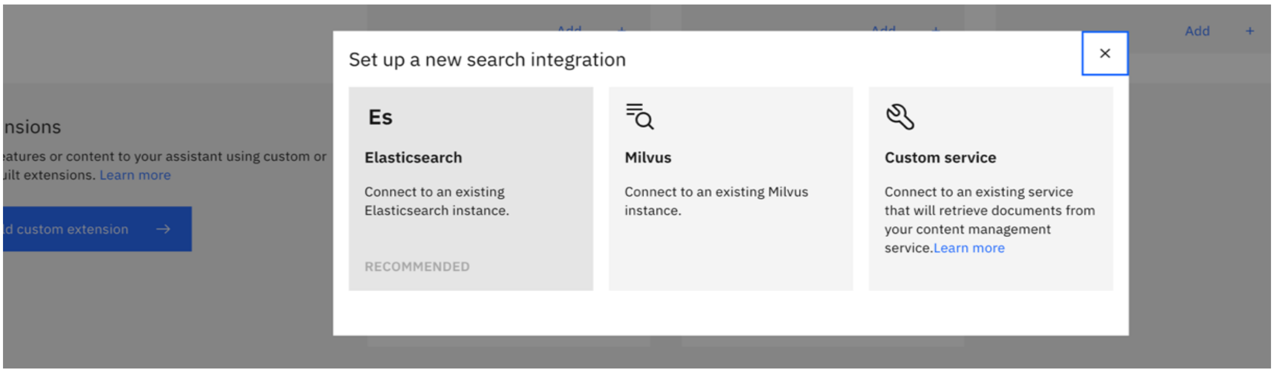
9.1 Watsonx data extension (Milvus)

9.1.1 From Left Navigation, select *Integrations* option.



9.1.2 Look for *Search* → Click on *Add* → select *Milvus*





9.1.3 Provide Milvus service details and Click Next

GRPC host: →Copy from Startup.txt

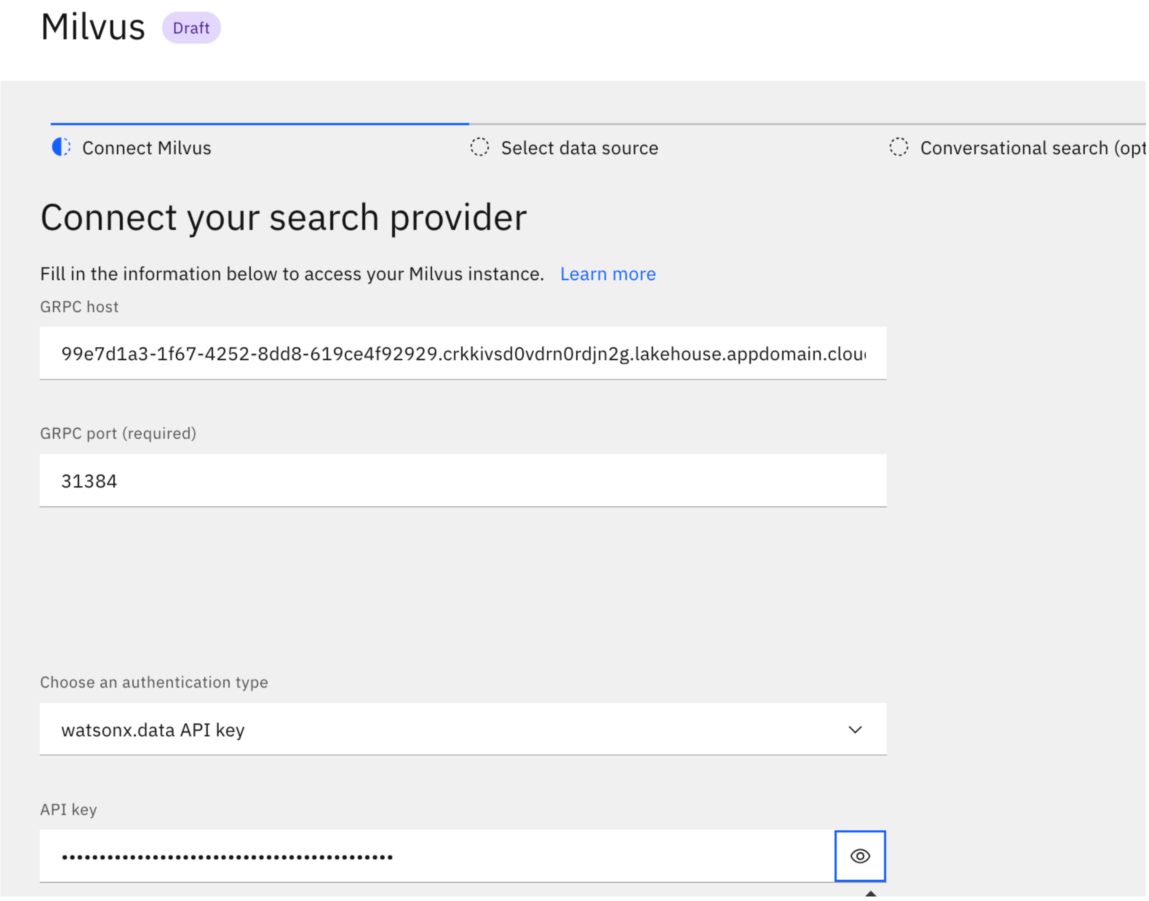
GRPC port: → Copy from Startup.txt

API key: → Copy from Startup.txt

In case it shows to enter user name and password, use below details

Username : ibmlhapikey

Password: API Key



9.1.4 Select *Database* → as Default

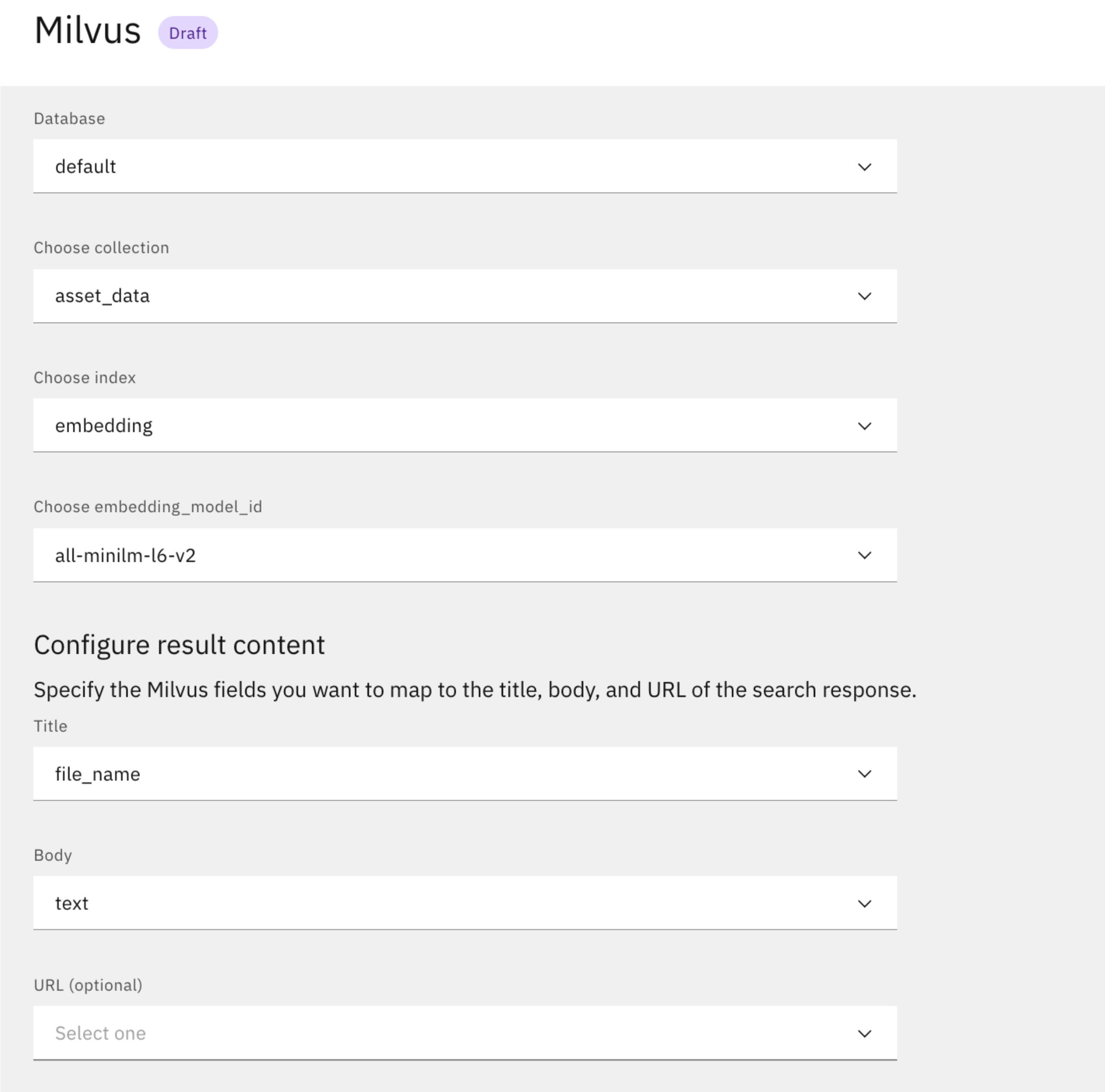
Collection name → asset_data

Choose index → *embedding*

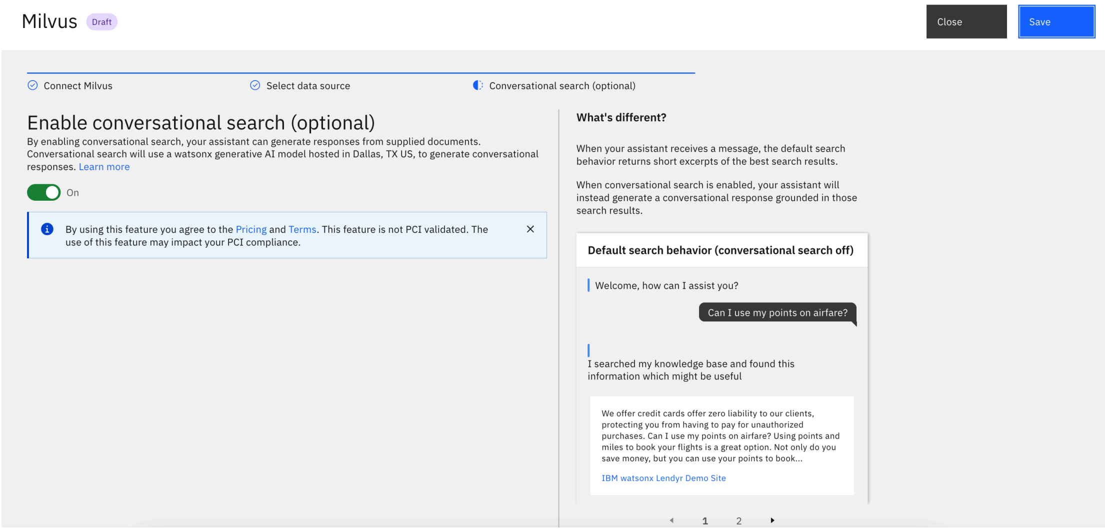
Choose embedding_model_Id → *all-minilm-l6-v2*

Title -> *file_name*

Body -> *text*



9.1.5 Click Next

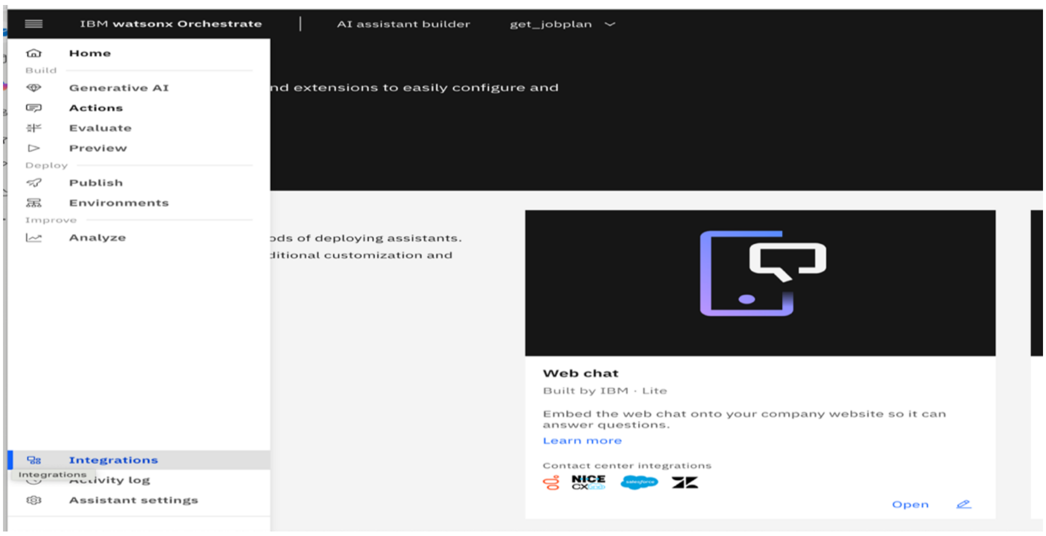


9.1.6 Click *Save* and *Close*.

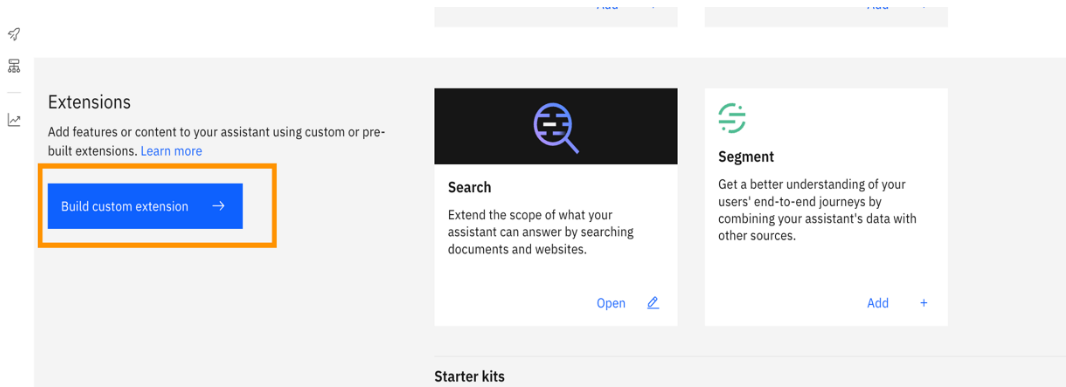
9.2 Maximo custom extension

9.2.1 Use **create_Jobplan.json** open Api file to add Maximo extension to the assistance

From Left Navigation, select *Integrations* option.

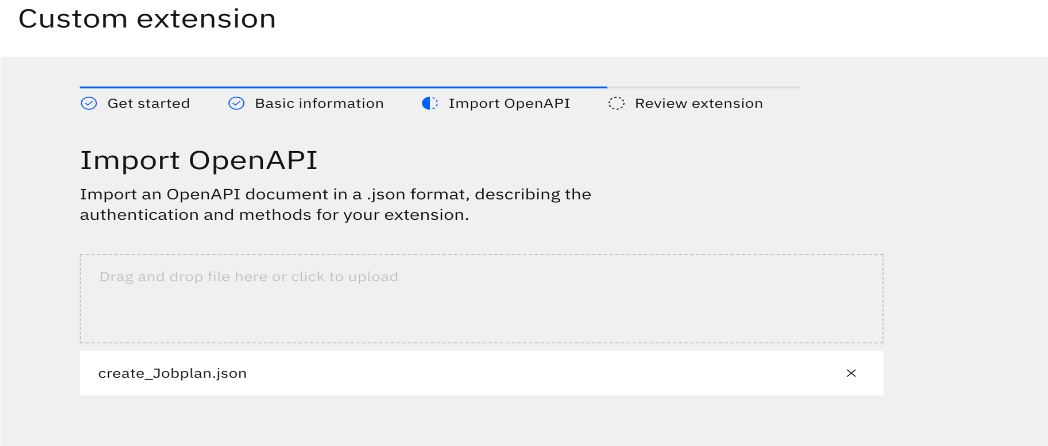


9.2.2 Click on *Build custom extension and click Next*

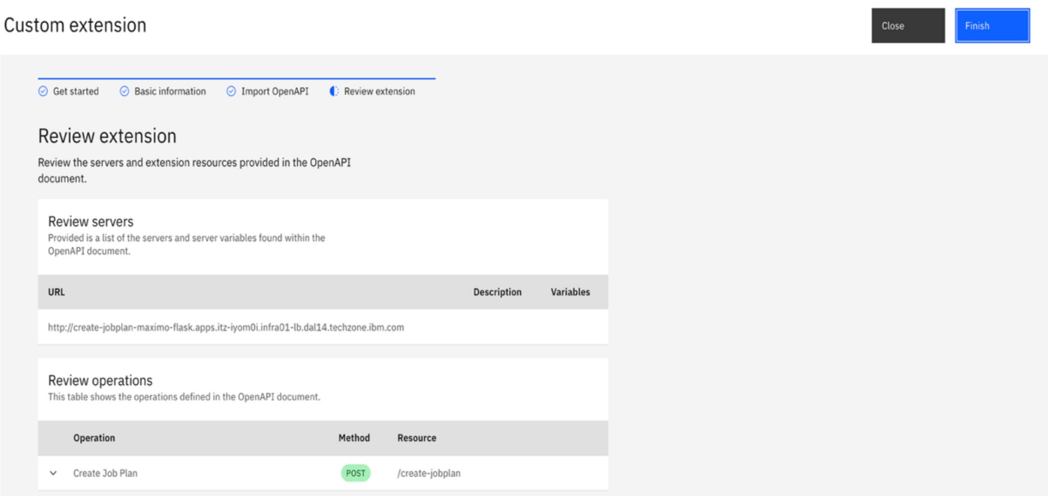


9.2.3 Enter Extension name as “maximo” and click next.

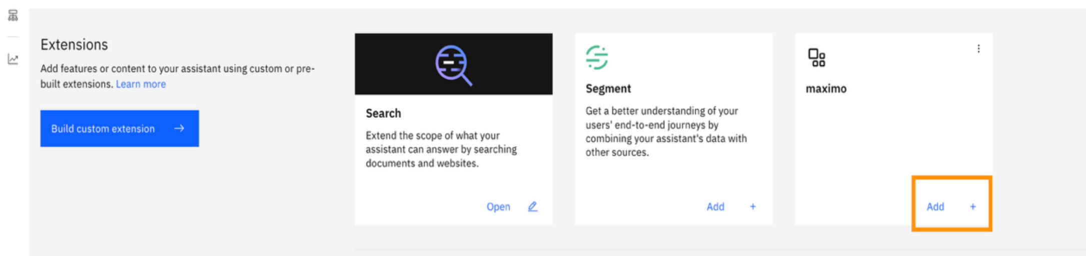
9.2.4 *Import open API* file “create_jobplan.json” and Click Next.



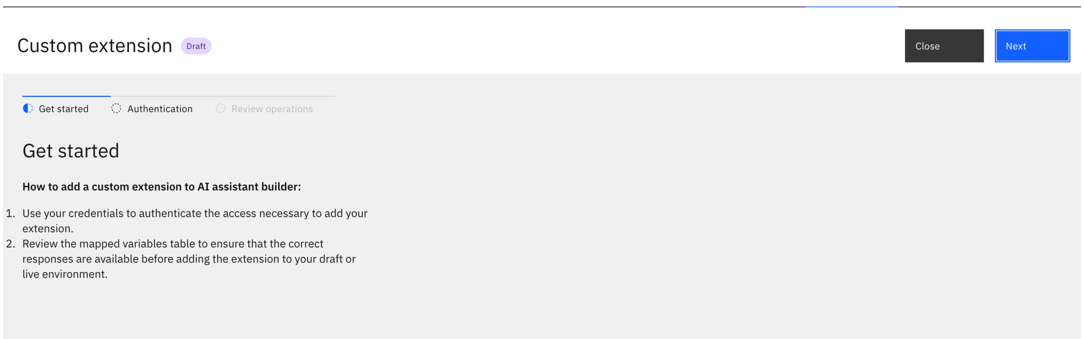
9.2.5 Review and *finish*.



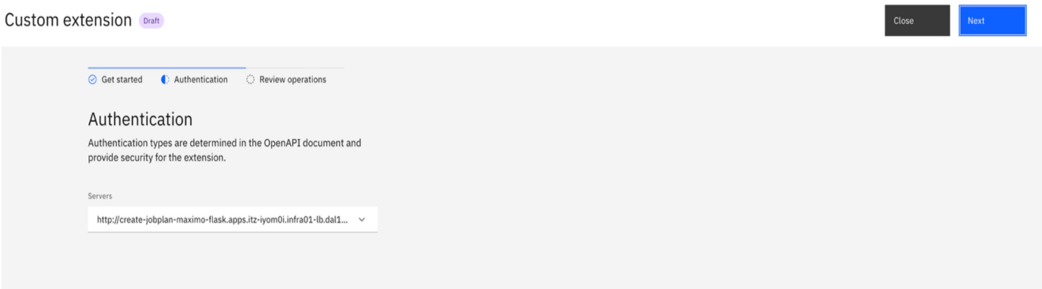
9.2.6 Then click on *Add*



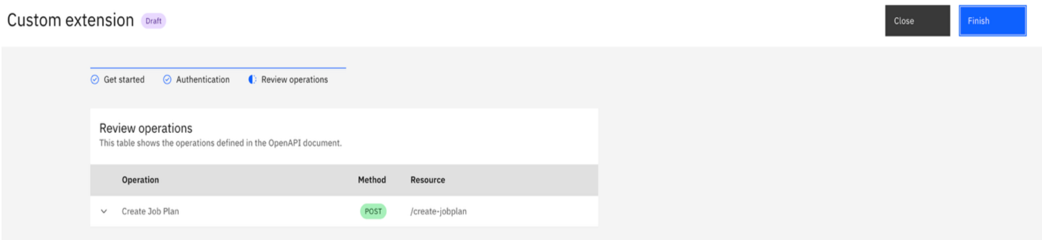
9.2.7 In Custom extension, click Next.



9.2.8 Under *Authentication* you can see the Maximo URL (Same URL will be present in the open API file)

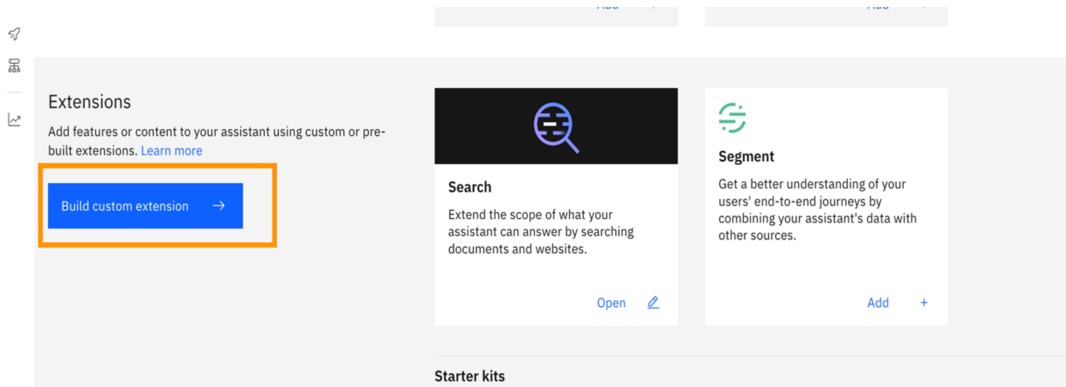


9.2.9 Click next & *finish*.

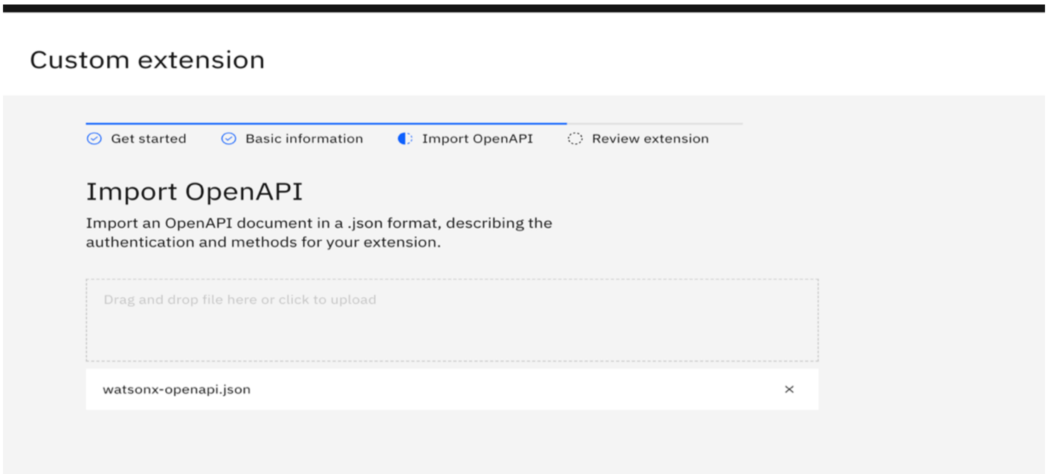


9.3 Watsonx AI integration

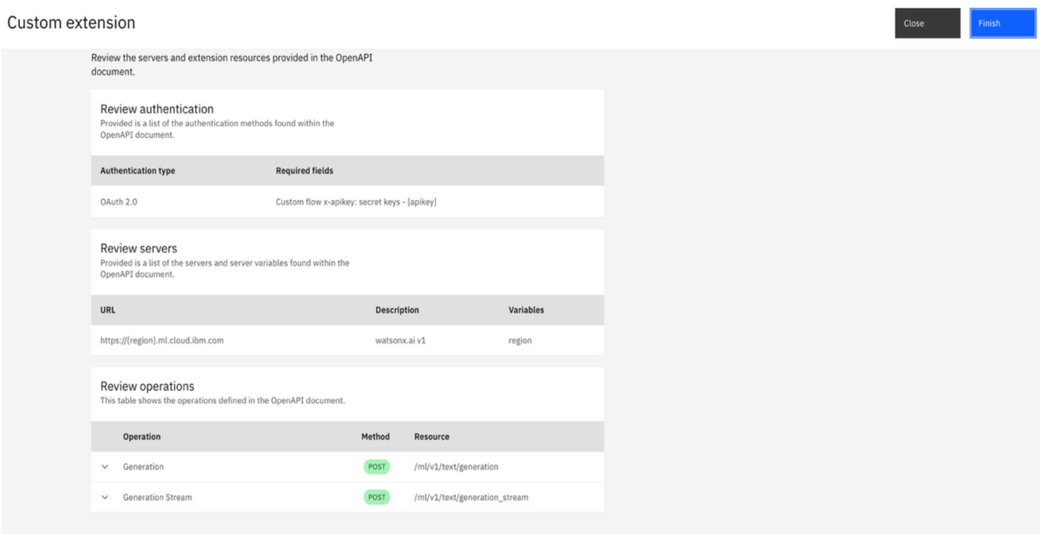
9.3.1 On the same Integrations page.
9.3.2 Click on *Build custom extension* again.



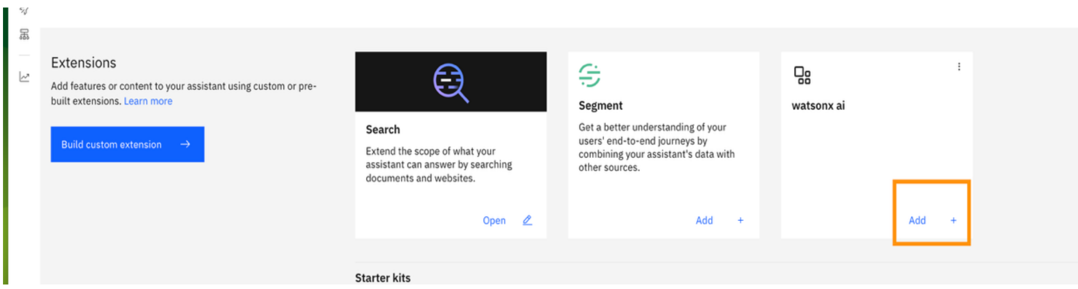
9.3.3 Enter Extension name as "watsonx ai"
9.3.4 In Import OpenAPI, select "**watsonx-openapi.json**" and click Next.



9.3.5 Review and click on *Finish*



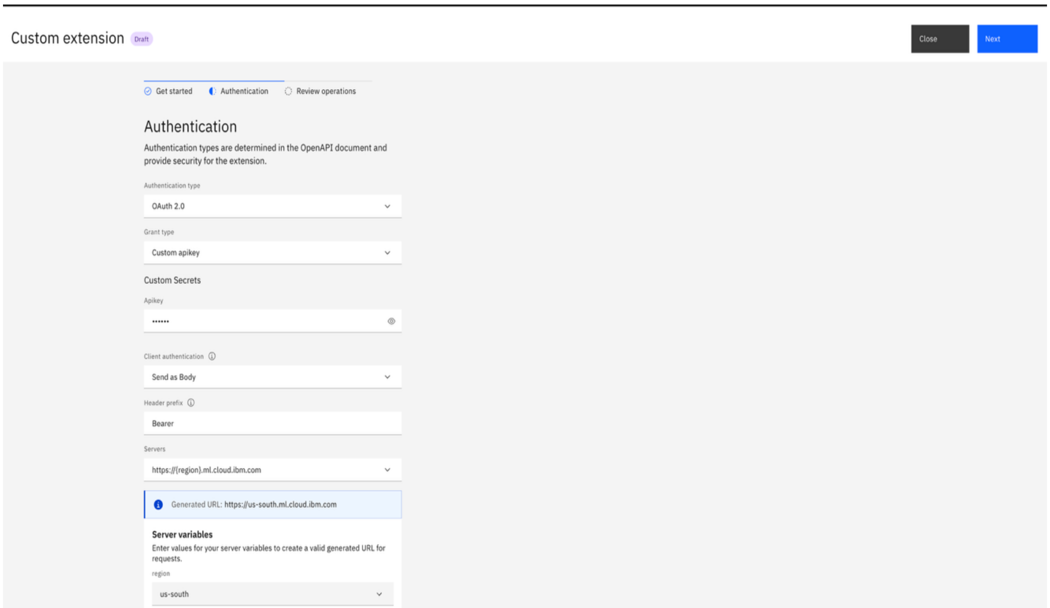
9.3.6 Click on *Add*



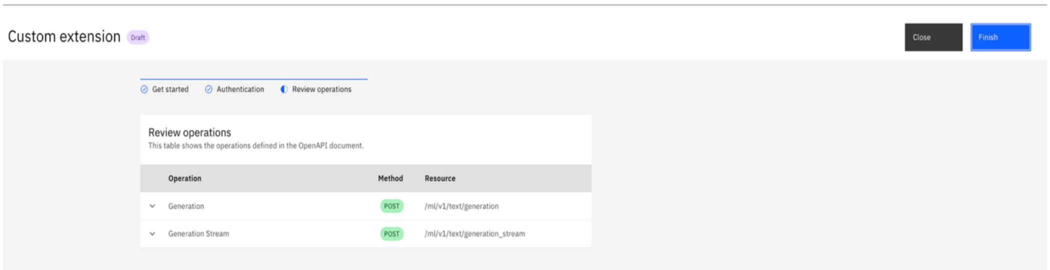
9.3.7 In Authentication screen,

Select/Provide below fields

a. Authentication type: OAuth 2.0
b. Grant type: Custom apikey
c. Api key: provide watsonx.ai apikey
d. Client authentication: sent as body



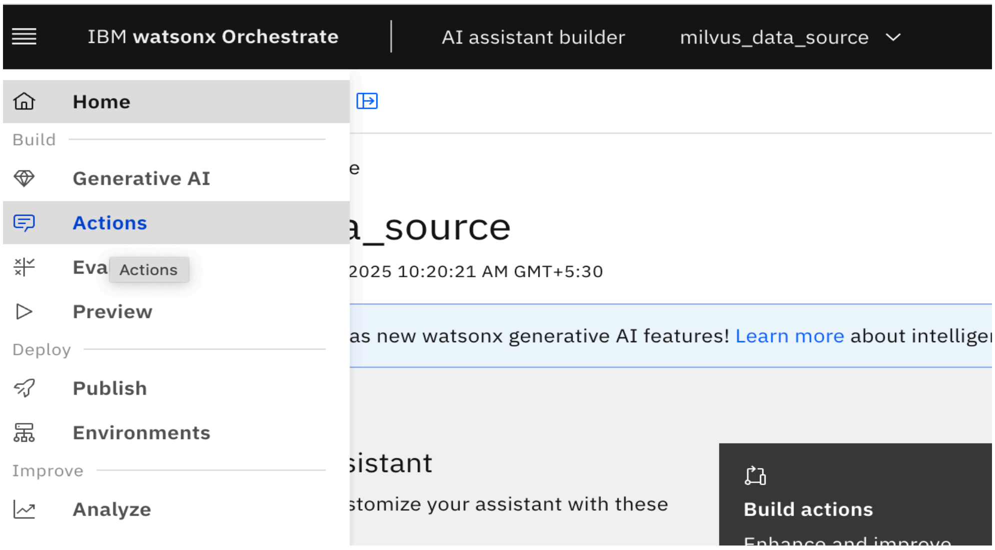
9.3.8 Click on next, Review & *Finish*



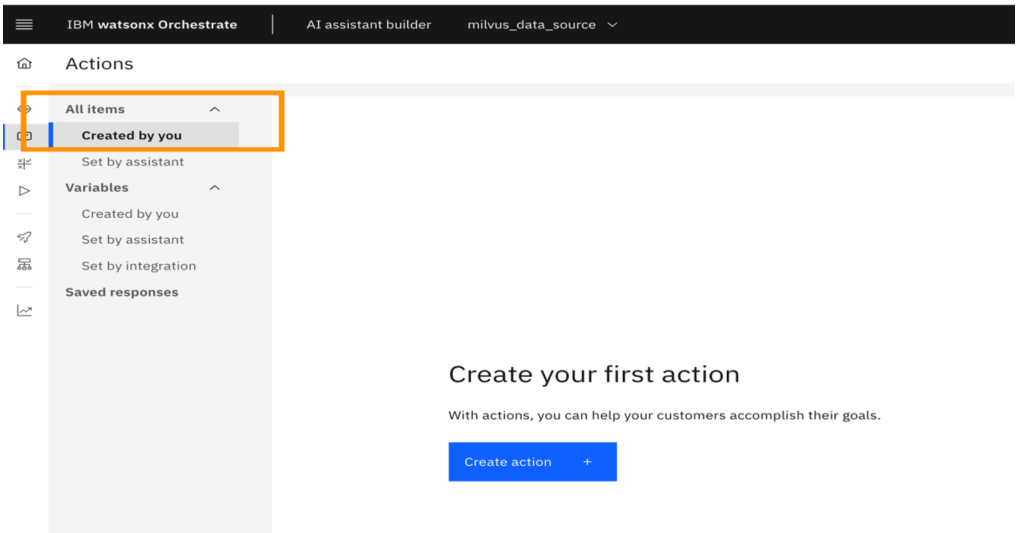
10. Import Actions

10.1Use the file actions.json file

10.1.1 Select *Actions* form the left side panel

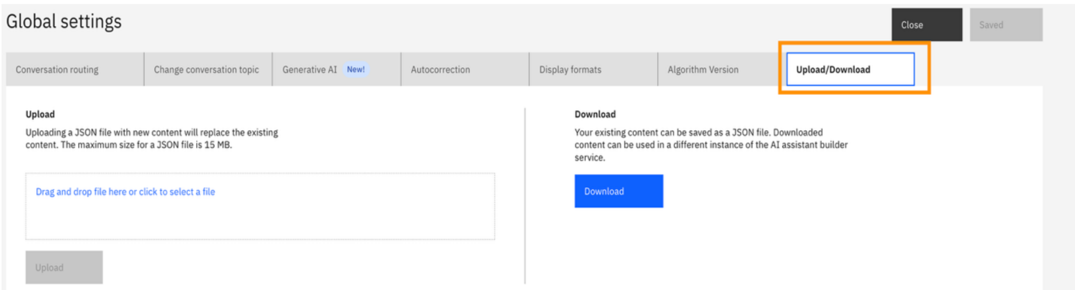


10.1.2 Click on *Created by You*

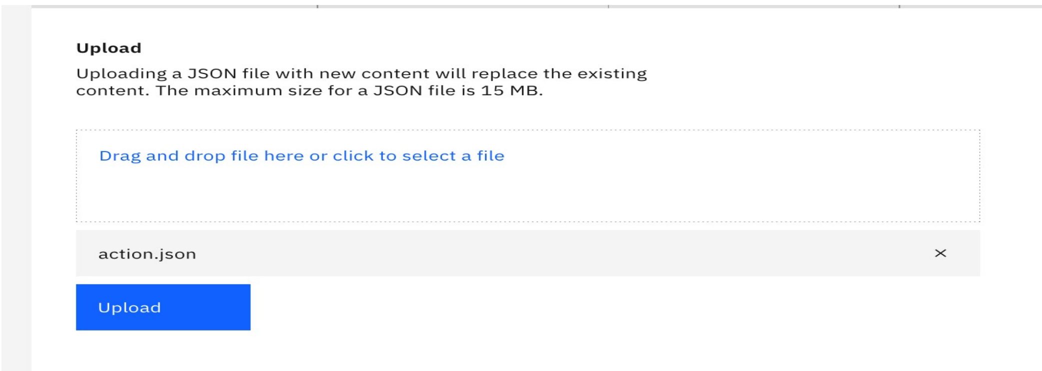


10.1.3 Select *Global settings* on right top corner

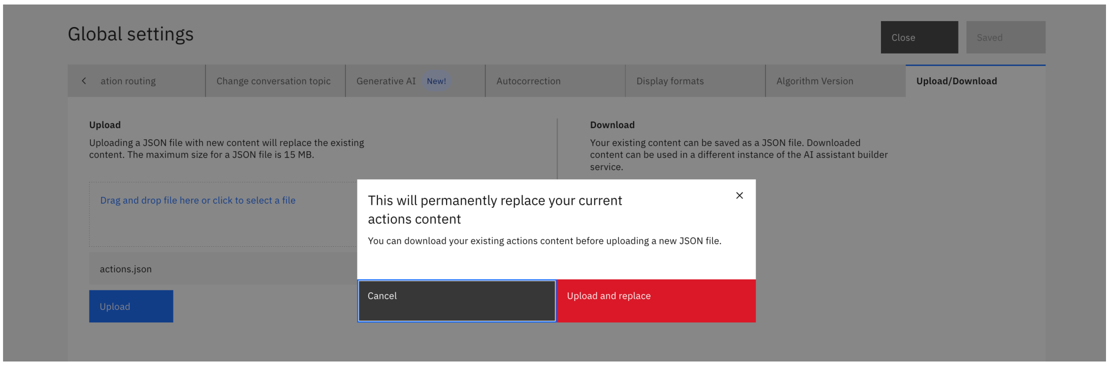
10.1.4 Navigate to *Upload/Download* option



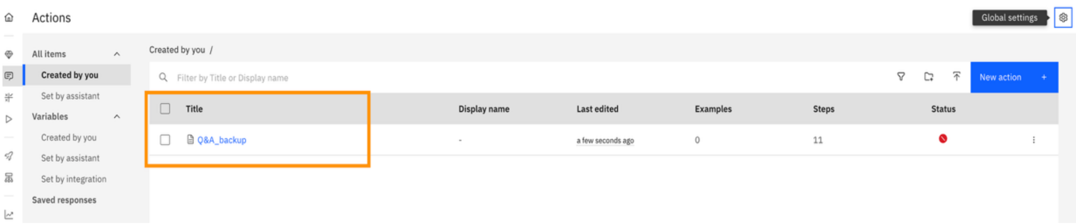
10.1.5 Click on *Drag and drop file here* option & select "actions.json" file →Click on *Upload*



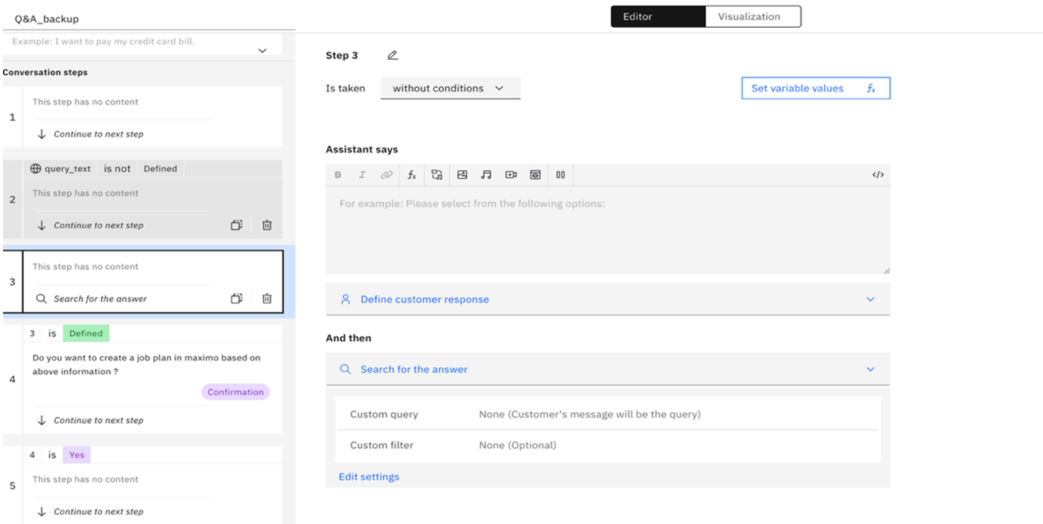
10.1.6 Select *Upload and replace* and Close the Global settings.



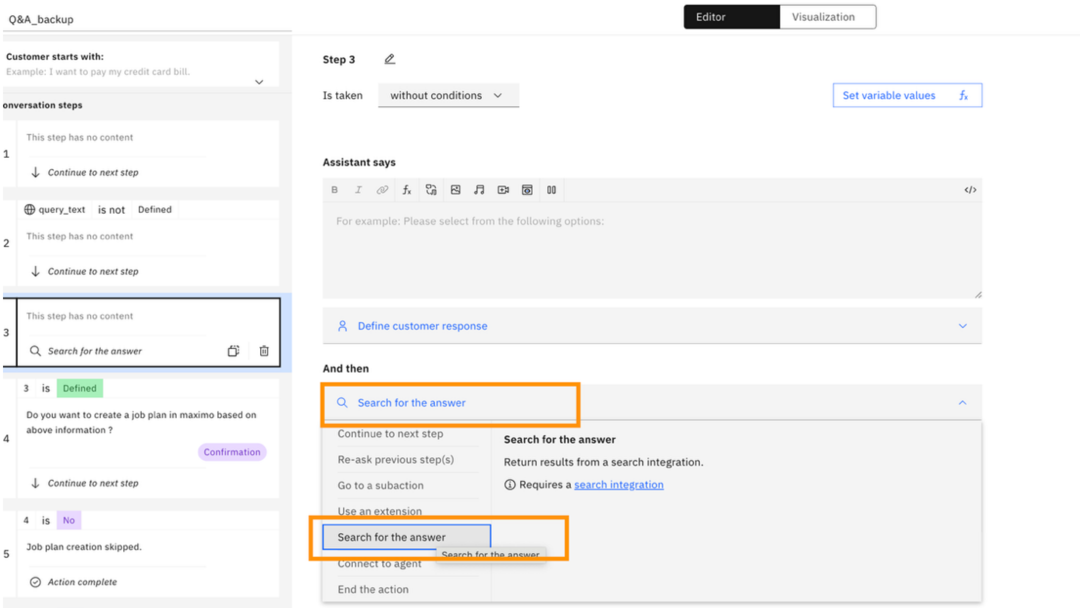
10.1.7 Click on *Q&A_backup* title



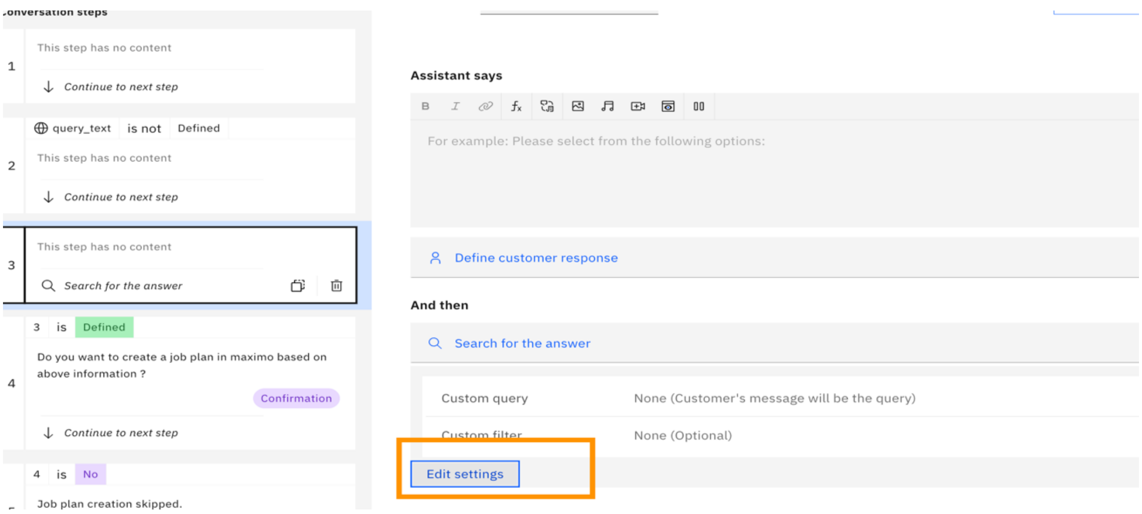
10.1.8 Click on Step3 →Make sure it is having *Search for the answer*

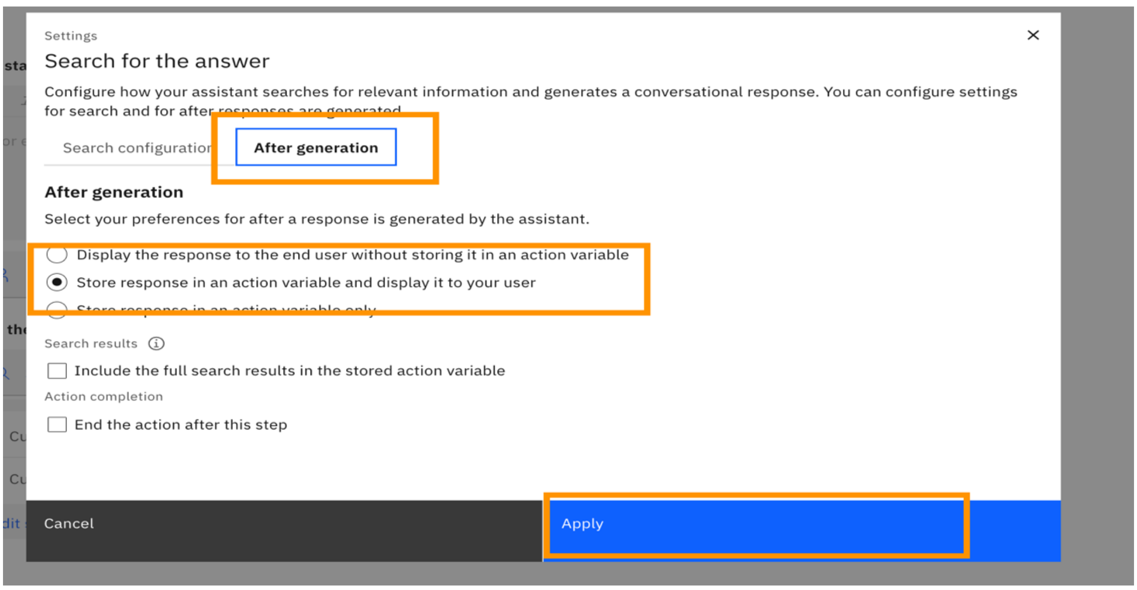


10.1.9   if not their select *Search for the answer*

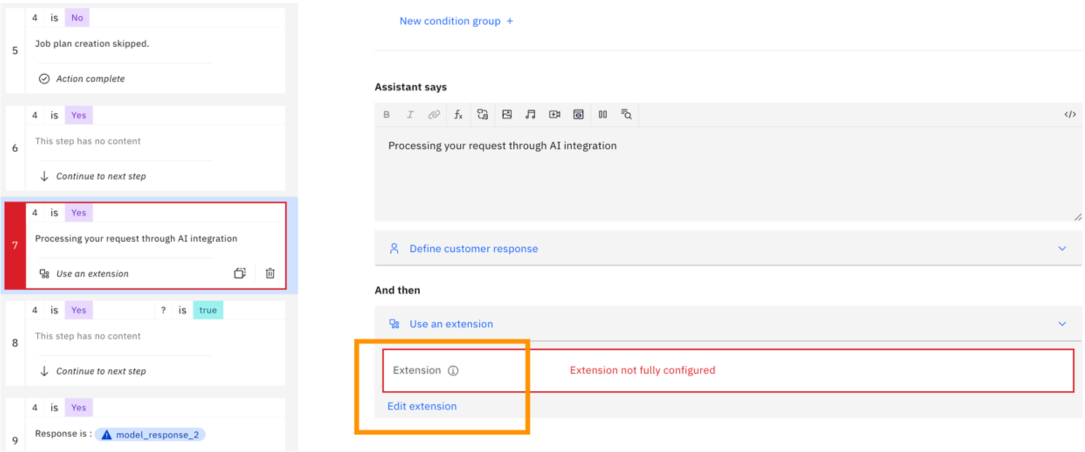


10.1.10 Click on *Edit settings* → Click on *After generation* → select 2nd option (highlighted in image) →Click on *Apply*





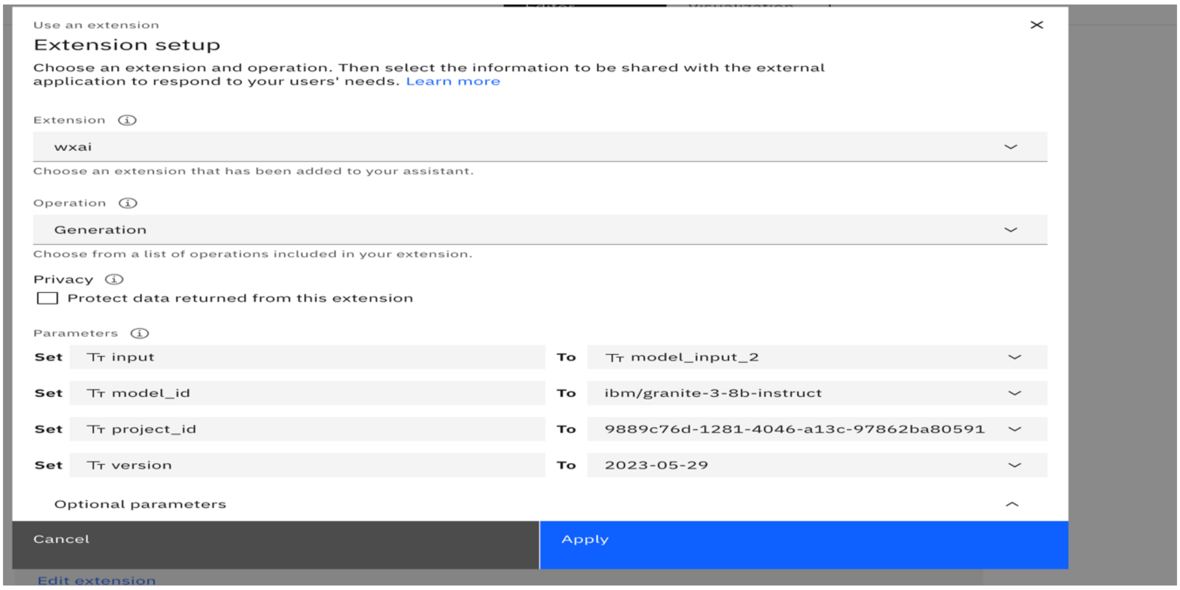
10.1.11 Select step 7 and click on *Edit extension* and use watsonx.ai extension.

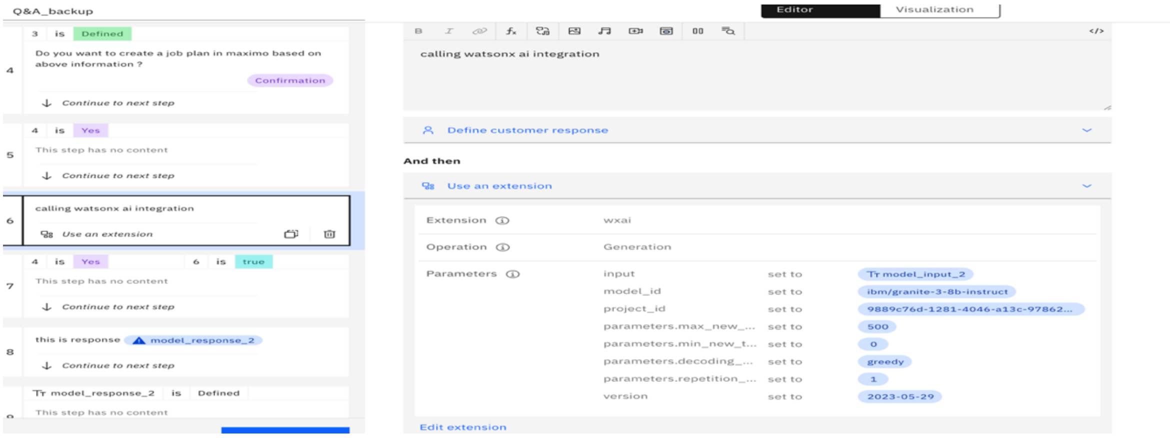


10.1.12 Update below fields → Click on *Apply*

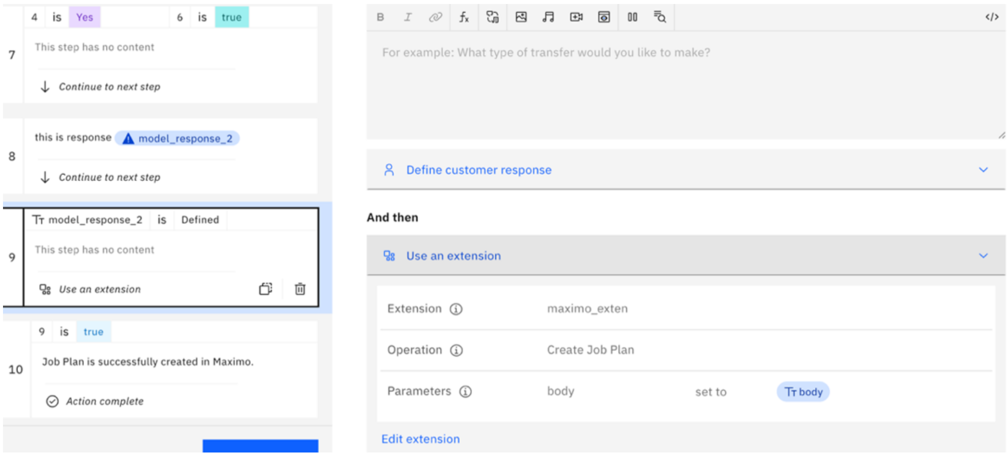
Edit the details if it appears different to below options.

Field name	Fetching from
input	Previous step output (ex: model_input_2)
Project_id	*project id from watsonx.ai → step 4.7*
Model_id	*ibm/granite-3-8b-instruct*
Version	2023-05-29
Max_tokens	500 & above (based on generated answer)
Min_tokens	0



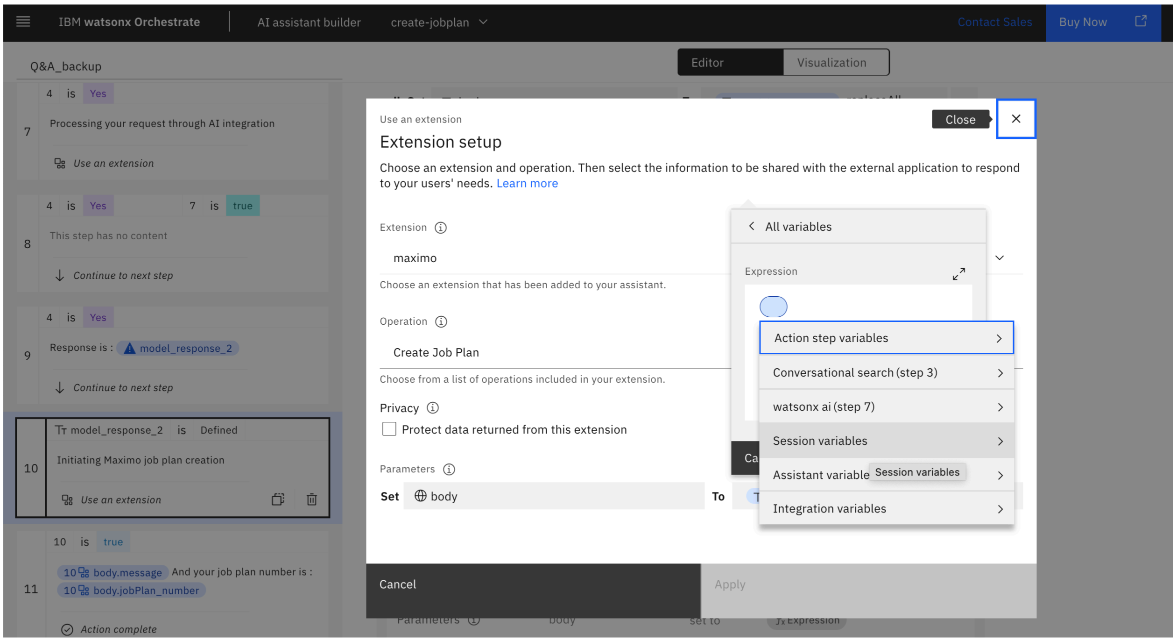


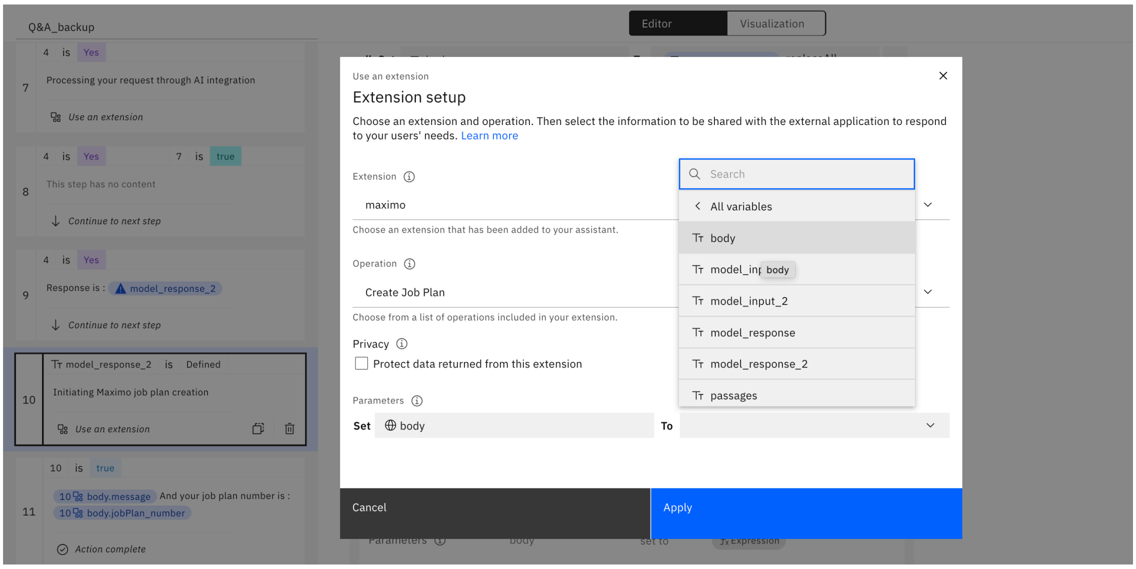
10.1.13 Select step 10 and click on *Edit extension* and use Maximo extension →provide *Parameters* value (previous step output in the actions (ex:body)



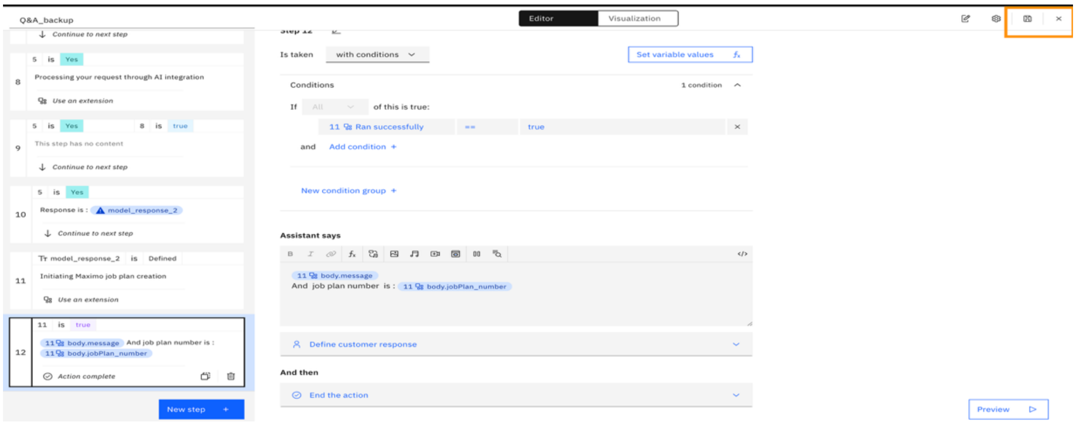
10.1.14 In Previous step, if Parameters is not set as Session variables (body) then follow this step.

- Click on Edit Extension
- Enter *Dollar ($) sign* and select *Session variables* -> and select *body* -> apply

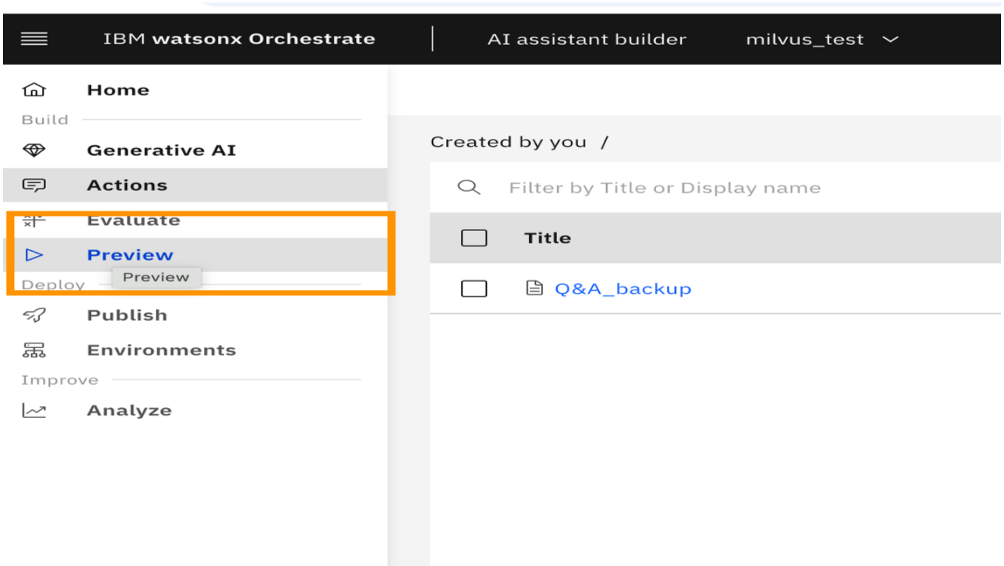




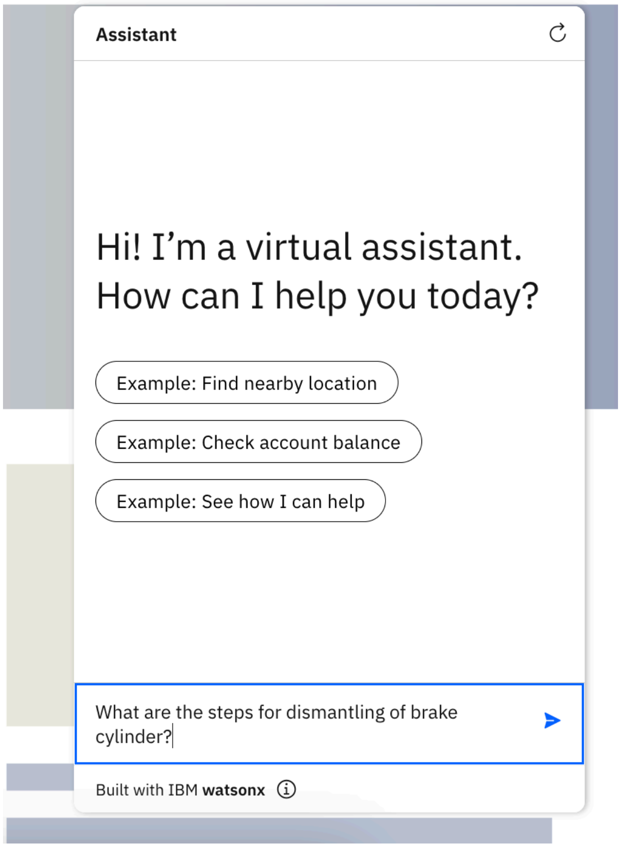
10.1.15 *Save* the actions → Click on *Close*



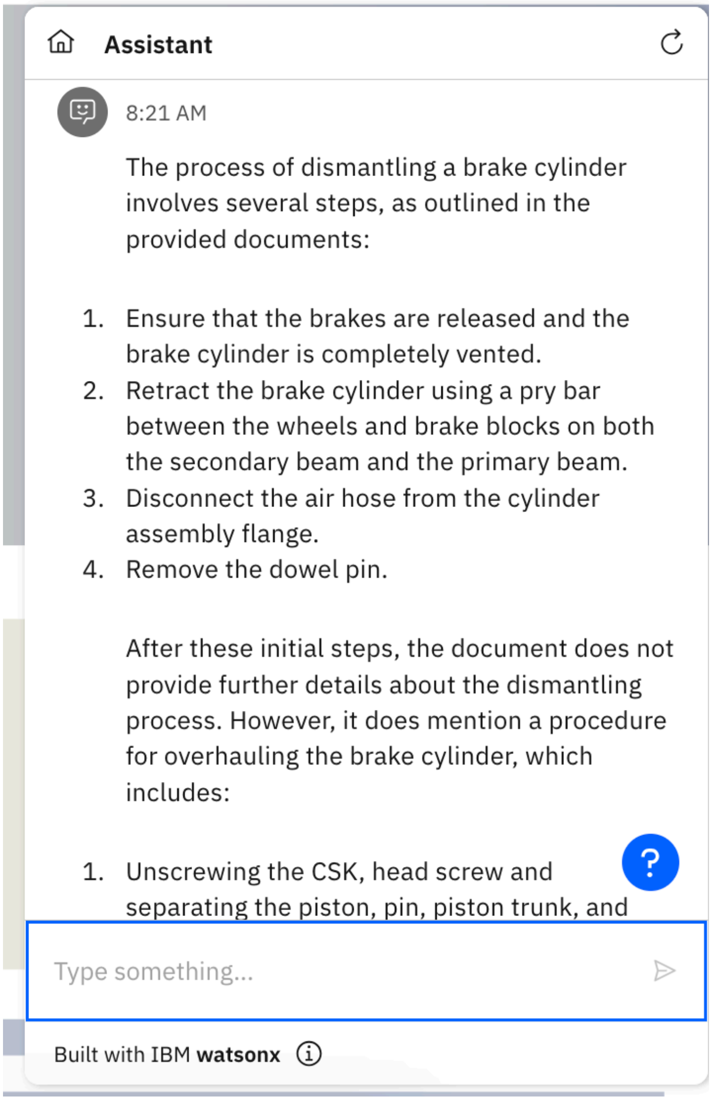
10.1.16 From *left* Navigation Click on *Preview*

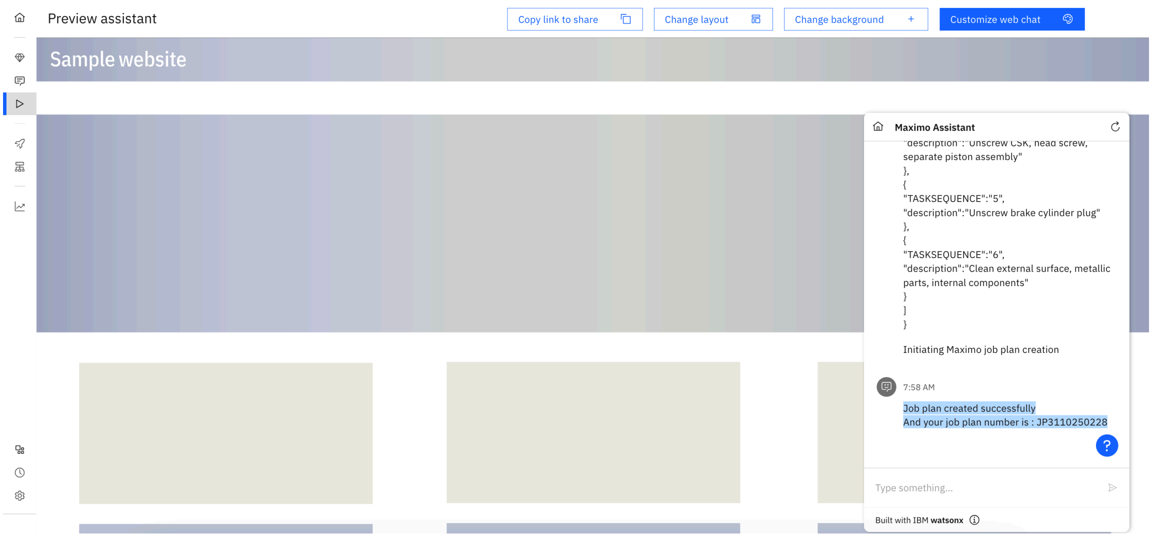


10.1.17 Ask Question like - What are the steps for dismantling of brake cylinder?



10.1.18 Answer will be generated and displayed on the interface.


Assistant
8:21 AM
The process of dismantling a brake cylinder involves several steps, as outlined in the provided documents:
1. Ensure that the brakes are released and the brake cylinder is completely vented.
2. Retract the brake cylinder using a pry bar between the wheels and brake blocks on both the secondary beam and the primary beam.
3. Disconnect the air hose from the cylinder assembly flange.
4. Remove the dowel pin.
After these initial steps, the document does not provide further details about the dismantling process. However, it does mention a procedure for overhauling the brake cylinder, which includes:
1. Unscrewing the CSK, head screw and separating the piston, pin, piston trunk, and
Type something...
Built with IBM watsonx


Preview assistant
Copy link to share
Change layout
Change background
Customize web chat
Sample website
Maximo Assistant
separate piston assembly"
},
{
"TASKSEQUENCE":"5",
"description":"Unscrew brake cylinder plug"
},
{
"TASKSEQUENCE":"6",
"description":"Clean external surface, metallic parts, internal components"
}
]
}
Initiating Maximo job plan creation
7:58 AM
Job plan created successfully
And your job plan number is : JP3110250228
Type something...
Built with IBM watsonx

11. Conclusion/Summary

The automated creation of job plans in Maximo from asset maintenance manuals streamlines a traditionally manual, error-prone process. By leveraging IBM Watsonx technologies—including Watsonx.ai, Watsonx.data with Milvus, and Watson Assistant—organizations can extract structured maintenance tasks from unstructured manuals and convert them into standardized job plans within Maximo.

This AI-powered workflow enables:

- Digitization of OEM maintenance procedures
- Faster and more accurate job plan creation
- Intelligent retrieval of relevant content using vector search (Milvus)
- Seamless integration with Maximo through custom APIs

Ultimately, this solution enhances operational efficiency, improves asset reliability, ensures compliance with manufacturer guidelines, and reduces the time and effort required by maintenance teams.

By adopting this approach, customers can modernize their maintenance strategy and unlock real value from existing technical documentation.